

Pontryagin Duality for Locally Compact Abelian Groups

A self-contained proof blueprint aligned with **Mathlib**

Prepared for autoformalization

Mathlib interface checked against v4.32.0 (July 2026)

Abstract

Let G be a Hausdorff locally compact abelian topological group and let

$$\widehat{G} = \text{Hom}_{\text{cts}}(G, \mathbb{T})$$

carry the compact–open topology. We give a complete proof that the evaluation map

$$\alpha_G : G \longrightarrow \widehat{\widehat{G}}, \quad \alpha_G(x)(\chi) = \chi(x),$$

is an isomorphism of topological groups. The proof is analytic: compact–open topology, Haar measure, the commutative Banach algebra $L^1(G)$, positive-type functions, the GNS construction, Bochner’s theorem, Fourier inversion, and Plancherel are developed in a dependency order suited to formalization. We then prove annihilator duality, duality for closed subgroups and quotients, extension of characters, exactness, compact/discrete duality, naturality, and the standard examples.

The target definition is exactly **Mathlib**’s multiplicative definition `PontryaginDual A := A ->_t* Circle`. Generic results that should be imported from the library—Haar existence, Riesz–Markov, Fubini–Tonelli, Banach–Alaoglu, Krein–Milman, and Stone–Weierstrass—are isolated in an explicit foundation interface. Every argument specific to Pontryagin duality is proved below; no form of biduality, character extension, or an LCA structure theorem is used as an input.

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1 Formalization contract and conventions

1.1 The mathematical category

Throughout the main proof, an *LCA group* means a type G equipped with a commutative group law and a Hausdorff locally compact topology for which multiplication and inversion

are continuous. We use multiplicative notation, including for $\mathbb{R}$ and $\mathbb{Z}$ when they occur as abstract groups. The circle is

$$\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$$

with multiplication and its usual topology. A *character* of G is a continuous group homomorphism $G \rightarrow \mathbb{T}$.

No countability, metrizability, compact generation, or second-countability hypothesis is made. Nets are used whenever sequential reasoning would introduce such a hypothesis. All measures are regular Borel (Radon) measures. Equalities of L^p functions are equalities almost everywhere; an equality between continuous functions obtained almost everywhere is upgraded to pointwise equality because Haar measure has full support.

1.2 The exact Mathlibinterface

In Mathlib v4.32.0 the relevant source file is `Mathlib/Topology/Algebra/PontryaginDual.lean`. Its central definition is:

```
def PontryaginDual := A ->_t* Circle
deriving TopologicalSpace
```

Here $A \rightarrow_t B$ is `ContinuousMonoidHomAB`; its topology is induced from the compact-open topology on continuous maps. The file already provides:

- (M1) a commutative group structure on `PontryaginDual A`;
- (M2) Hausdorff and topological-group instances;
- (M3) local compactness of `PontryaginDual G` when G is a locally compact topological group;
- (M4) compactness of the dual of a discrete group and discreteness of the dual of a compact group;
- (M5) the contravariant map

```
def PontryaginDual.map (f : A ->_t* B) :
  PontryaginDual B ->_t* PontryaginDual A :=
  f.compLeft Circle
```

together with `map_apply`, `map_one`, `map_comp`, `map_mul`, and `mapHom`.

The intended new definition and theorem have the following shape. Names are proposed, not asserted to exist already.

```
noncomputable def PontryaginDual.eval
  (G : Type*) [CommGroup G] [TopologicalSpace G]
  [IsTopologicalGroup G] [LocallyCompactSpace G] :
  G ->_t* PontryaginDual (PontryaginDual G) := ...

theorem PontryaginDual.eval_bijective
  [T2Space G] : Function.Bijective (PontryaginDual.eval G) := ...

noncomputable def PontryaginDual.toDoubleDual
  [T2Space G] :
  ContinuousMulEquiv G (PontryaginDual (PontryaginDual G)) := ...
```

The defining simp lemma should be

```
@[simp] theorem eval_apply (x : G) (chi : PontryaginDual G) :
  PontryaginDual.eval G x chi = chi x := rfl
```

and naturality should be exposed in a form that rewrites without category theory:

```
theorem map_map_eval (f : G ->_t* H) :
  ContinuousMonoidHom.comp
    (PontryaginDual.map (PontryaginDual.map f))
    (PontryaginDual.eval G)
  = ContinuousMonoidHom.comp (PontryaginDual.eval H) f := ...
```

1.3 Foundation interface

Calling the account “self-contained” means that every Pontryagin-specific construction and inference is present. Re-proving the foundations of measure theory and functional analysis inside the same document would obscure the formalization boundary. The proof uses precisely the following generic results, all stated here so that no hidden mathematical hypothesis is introduced.

Label	Imported statement
(F1)	A locally compact Hausdorff space is regular. If $x \in U$ with U open, there is $u \in C_c(X, [0, 1])$ with $u(x) = 1$ and $\text{supp } u \subseteq U$. Compact subsets have compact neighbourhoods.
(F2)	Every locally compact Hausdorff group has a nonzero left Haar measure, unique up to a positive scalar. Haar measure is Radon, finite on compact sets, positive on nonempty open sets, and has full support. On an abelian group it is also right invariant and invariant under inversion.
(F3)	Tonelli, Fubini, dominated convergence, monotone convergence, change of variables under a measure-preserving homeomorphism, density of $C_c(G)$ in $L^p(G)$ for $1 \leq p < \infty$, Hölder, and the Riesz representation theorem for Hilbert spaces.
(F4)	Riesz–Markov: positive linear functionals on $C_c(X)$ that are locally bounded are represented by unique Radon measures. Finite complex Radon measures are the dual of $C_0(X)$; their total variation is regular.
(F5)	Tychonoff’s theorem and the Arzelà–Ascoli theorem in compact–open form: a closed, pointwise relatively compact, equicontinuous family of continuous maps is compact on a locally compact domain.
(F6)	Banach–Alaoglu for dual balls; Krein–Milman for compact convex subsets of a locally convex space; Stone–Weierstrass for self-adjoint, point-separating subalgebras of $C_0(X)$ that vanish nowhere.
(F7)	The standard construction of quotient topological groups by closed normal subgroups, and completion of a positive semidefinite sesquilinear space to a Hilbert space.
(F8)	Unitization and the spectral-radius bound for Banach algebras, and the spectral theorem (equivalently, continuous functional calculus and spectral projections) for bounded normal operators on a complex Hilbert space.

The current library contains the principal ingredients of (F1)–(F8), though some glue lemmas in exactly the stated generality may need to be added. In particular, the proof

must not replace nets by sequences or invoke an L^1 duality theorem with an unstated σ -finiteness assumption.

1.4 Fourier conventions

Fix a Haar measure μ_G on G . For $a \in G$ set

$$(L_a f)(x) = f(a^{-1}x), \quad f^*(x) = \overline{f(x^{-1})}, \quad (f * g)(x) = \int_G f(y)g(y^{-1}x) d\mu_G(y).$$

For $f \in L^1(G)$ define

$$\widehat{f}(\chi) = \int_G f(x) \overline{\chi(x)} d\mu_G(x). \quad (1)$$

For a finite complex measure σ on $\widehat{G}$, define its inverse Fourier–Stieltjes transform by

$$\check{\sigma}(x) = \int_{\widehat{G}} \chi(x) d\sigma(\chi). \quad (2)$$

Thus the desired inversion formula is

$$f(x) = \int_{\widehat{G}} \widehat{f}(\chi) \chi(x) d\mu_{\widehat{G}}(\chi). \quad (3)$$

These conventions give

$$\widehat{f * g} = \widehat{f} \widehat{g}, \quad \widehat{f^*} = \overline{\widehat{f}}, \quad \widehat{L_a f}(\chi) = \overline{\chi(a)} \widehat{f}(\chi). \quad (4)$$

All later conjugations and inverses are fixed by these three identities.

2 The circle and the compact–open character group

2.1 Small arcs and the no-small-subgroups lemma

For $0 < r < \pi$ put

$$A_r = \{z \in \mathbb{T} : |\arg z| < r\},$$

where $\arg z \in (-\pi, \pi]$ is the principal argument.

Lemma 2.1 (No small subgroups). *If $z^n \in A_{\pi/2}$ for every $n \in \mathbb{Z}$, then $z = 1$. Consequently the only subgroup of $\mathbb{T}$ contained in $A_{\pi/2}$ is the trivial subgroup.*

Proof. Write $z = e^{i\theta}$ with $\theta \in (-\pi, \pi]$. If $\theta \neq 0$, put $t = |\theta| \in (0, \pi]$ and take $n = \lceil \pi/(2t) \rceil$. If $t > \pi/2$, then $n = 1$; if $t \leq \pi/2$, then

$$\frac{\pi}{2} \leq nt < \frac{\pi}{2} + t \leq \pi.$$

In either case $nt \in [\pi/2, \pi]$. Therefore the principal argument of z^n has absolute value at least $\pi/2$, contradicting $z^n \in A_{\pi/2}$. Hence $\theta = 0$. In `Mathlib` this is packaged as `Circle.eq_one_of_forall_pow_mem_centeredArc_pi_div_two`. $\square$

The following quantitative form is what compactness and equicontinuity arguments use.

Lemma 2.2 (Power control). *Let W be a neighbourhood of 1 in $\mathbb{T}$. There is $m \geq 1$ such that, for every $z \in \mathbb{T}$,*

$$z, z^2, \dots, z^m \in \overline{A_{\pi/4}} \implies z \in W.$$

Proof. If not, for every m choose $z_m \notin W$ whose first m powers lie in the closed arc. Compactness of $\mathbb{T}$ gives a convergent subnet $z_m \rightarrow z \notin W$ after first replacing W by a neighbourhood with closed closure contained in the original W . For each fixed n , eventually $m \geq n$, so $z_m^n \in \overline{A_{\pi/4}}$ and hence $z^n \in \overline{A_{\pi/4}} \subset A_{\pi/2}$. The same holds for negative powers. [Theorem 2.1](#) gives $z = 1$, contradicting $z \notin W$. $\square$

Formalization note. The local-compactness instance in the current source uses the explicit basis $A_{\pi/2^{n+1}}$ and a one-step square-root lemma instead of [theorem 2.2](#). Reusing that exact basis will substantially shorten Lean proofs.

2.2 The compact–open topology

For compact $K \subseteq G$ and open $U \subseteq \mathbb{T}$ write

$$[K, U] = \{f \in C(G, \mathbb{T}) : f(K) \subseteq U\}.$$

The compact–open topology on $C(G, \mathbb{T})$ is generated by these sets, and $\widehat{G}$ has the induced topology. At the trivial character it is enough to take sets $[K, U]$ with U a neighbourhood of 1.

Lemma 2.3 (Neighbourhood basis). *The sets*

$$W(K, U) = \{\chi \in \widehat{G} : \chi(K) \subseteq U\},$$

where K is compact and U is a neighbourhood of 1 in $\mathbb{T}$, form a neighbourhood basis at $1 \in \widehat{G}$.

Proof. A basic neighbourhood of the constant map 1 is a finite intersection $\bigcap_j [K_j, U_j]$. Put $K = \bigcup_j K_j$ and choose a neighbourhood $U \subseteq \bigcap_j U_j$ of 1. Then $W(K, U)$ is contained in the intersection. Conversely each $W(K, U)$ is the intersection of $\widehat{G}$ with a subbasic compact–open set, hence is open in $\widehat{G}$. $\square$

Proposition 2.4 (Joint evaluation). *If G is locally compact Hausdorff, then*

$$\text{ev} : \widehat{G} \times G \longrightarrow \mathbb{T}, \quad (\chi, x) \longmapsto \chi(x)$$

is continuous.

Proof. Fix (χ_0, x_0) and a neighbourhood O of $\chi_0(x_0)$. Choose a symmetric neighbourhood V of 1 with $VV\chi_0(x_0) \subseteq O$. By continuity of χ_0 , there is a neighbourhood N of x_0 on which $\chi_0(x)\chi_0(x_0)^{-1} \in V$. By local compactness shrink N so that its closure K is compact. If $\chi\chi_0^{-1} \in W(K, V)$ and $x \in N$, then

$$\chi(x)\chi_0(x_0)^{-1} = (\chi\chi_0^{-1})(x)\chi_0(x)\chi_0(x_0)^{-1} \in VV.$$

Thus $W(K, V)\chi_0 \times N$ is mapped into O . $\square$

Formalization note. This is already abstracted by the `ContinuousEval` instance for continuous monoid homomorphisms under `LocallyCompactPair`. Currying uses the existing lemma `continuous_of_continuous_uncurry` in the `ContinuousMonoidHom` namespace.

2.3 The dual is again locally compact abelian

Theorem 2.5. *If G is an LCA group, then $\widehat{G}$, with pointwise multiplication and the compact–open topology, is an LCA group.*

Proof. Pointwise multiplication and inversion preserve characters. Their continuity follows from the induced compact–open topology and continuity of the corresponding pointwise operations on $C(G, \mathbb{T})$. Since $\mathbb{T}$ is Hausdorff and two distinct characters differ at some point, the compact–open function space, and hence $\widehat{G}$, is Hausdorff.

It remains to prove local compactness. Choose a compact neighbourhood K of 1 in G and an arc A_r whose closure is contained in $A_{\pi/4}$. Let

$$E = \left\{ \chi \in \widehat{G} : \chi(K) \subseteq \overline{A_r} \right\}.$$

This family is closed in $\widehat{G}$ and pointwise relatively compact because $\mathbb{T}$ is compact. It is equicontinuous at 1. Indeed, given a neighbourhood W of 1 in $\mathbb{T}$, take m from [theorem 2.2](#). Choose a neighbourhood V of 1 in G such that $V^j \subseteq K$ for $1 \leq j \leq m$. For $v \in V$ and $\chi \in E$, all powers $\chi(v)^j = \chi(v^j)$ lie in $\overline{A_r}$, so $\chi(v) \in W$. Translation and the homomorphism identity turn equicontinuity at 1 into equicontinuity everywhere. Arzelà–Ascoli now makes E compact. Finally $W(K, A_r)$ is an open neighbourhood of 1 whose closure is contained in E . Thus $\widehat{G}$ has a compact neighbourhood of its identity. $\square$

Formalization note. The theorem is already an instance. Its proof in v4.32.0 invokes `ContinuousMonoidHom.locallyCompactSpace_of_hasBasis` with the centred arcs $A_{\pi/2^{n+1}}$; it ultimately uses Arzelà–Ascoli. The argument above records the mathematical invariants that must remain visible if that implementation is refactored.

Proposition 2.6 (Compact/discrete directions). *(a) If G is discrete, then $\widehat{G}$ is compact.*

(b) If G is compact, then $\widehat{G}$ is discrete.

Proof. If G is discrete, every homomorphism $G \rightarrow \mathbb{T}$ is continuous. The set of homomorphisms is closed in the compact product $\mathbb{T}^G$, because the equations $f(1) = 1$ and $f(xy) = f(x)f(y)$ are closed conditions. Compact subsets of a discrete space are finite, so the product and compact–open topologies agree on this closed set.

If G is compact, $W(G, A_{\pi/2})$ contains only the trivial character: the image of any character in this set is a subgroup of $\mathbb{T}$ contained in $A_{\pi/2}$, hence is trivial by [theorem 2.1](#). The singleton $\{1\}$ is therefore open, and translations show that every singleton is open. $\square$

3 Functoriality and the evaluation morphism

For a continuous homomorphism $f : G \rightarrow H$, define

$$\widehat{f} : \widehat{H} \longrightarrow \widehat{G}, \quad (\widehat{f})(\chi) = \chi \circ f.$$

Continuity follows from continuity of precomposition in the compact–open topology. Direct evaluation proves

$$\widehat{\text{id}_G} = \text{id}_{\widehat{G}}, \quad \widehat{(g \circ f)} = (\widehat{f}) \circ (\widehat{g}).$$

This is exactly `PontryaginDual.map` and `map_comp`.

Proposition 3.1 (Evaluation morphism). *For an LCA group G , the formula*

$$\alpha_G(x)(\chi) = \chi(x)$$

defines a continuous homomorphism $\alpha_G : G \rightarrow \widehat{\widehat{G}}$.

Proof. For fixed x , evaluation is multiplicative in χ , so $\alpha_G(x)$ is a group homomorphism. It is continuous by continuity of point evaluation on the compact–open space. The identity $\alpha_G(xy)(\chi) = \chi(xy) = \chi(x)\chi(y)$ makes α_G a homomorphism. Finally the uncurried map $(x, \chi) \mapsto \chi(x)$ is continuous by [theorem 2.4](#); the compact–open exponential law, in the precise form `continuous_of_continuous_uncurry`, says that its curry α_G is continuous. $\square$

Proposition 3.2 (Naturality and the triangle identity). *For every continuous homomorphism $f : G \rightarrow H$,*

$$\widehat{\widehat{f}} \circ \alpha_G = \alpha_H \circ f. \quad (5)$$

Moreover

$$(\widehat{\alpha_G}) \circ \alpha_{\widehat{G}} = \text{id}_{\widehat{G}}. \quad (6)$$

Proof. For $x \in G$ and $\psi \in \widehat{H}$,

$$(\widehat{\widehat{f}}(\alpha_G(x)))(\psi) = \alpha_G(x)(\psi \circ f) = \psi(f(x)) = \alpha_H(f(x))(\psi).$$

For $\chi \in \widehat{\widehat{G}}$ and $x \in G$,

$$((\widehat{\alpha_G})(\alpha_{\widehat{G}}(\chi)))(x) = \alpha_{\widehat{G}}(\chi)(\alpha_G(x)) = \alpha_G(x)(\chi) = \chi(x).$$

Extensionality of continuous homomorphisms proves both identities. $\square$

4 Haar measure, convolution, and approximate identities

Fix an LCA group G and a nonzero Haar measure μ_G . Commutativity implies that left translation, right translation, and inversion preserve μ_G .

Lemma 4.1 (Normalized compactly supported bumps). *For every identity neighbourhood U there is $h_U \in C_c(G)$ such that*

$$h_U \geq 0, \quad \text{supp } h_U \subseteq U, \quad \int_G h_U d\mu_G = 1.$$

The support may be required to lie in any prescribed relatively compact identity neighbourhood contained in U .

Proof. Choose a nonempty open set V with compact closure contained in U . By (F1), choose a nonzero $u \in C_c(G, [0, 1])$ supported in U and positive at a point of V . Continuity makes u positive on a nonempty open set, so full support of Haar measure gives $0 < \int u < \infty$. Put $h_U = u / (\int u)$. $\square$

Order the identity neighbourhoods by reverse inclusion and make one choice h_U for each U . This is the approximate identity used below.

Lemma 4.2 (Continuity of translation). *For $1 \leq p < \infty$ and $f \in L^p(G)$,*

$$\|L_a f - f\|_p \longrightarrow 0 \quad (a \rightarrow 1).$$

Consequently $a \mapsto L_a f$ is continuous from G to $L^p(G)$.

Proof. First take $f \in C_c(G)$. For a in a fixed compact neighbourhood of 1, all $L_a f - f$ have support in one compact set. Uniform continuity of f on a slightly larger compact set gives $\|L_a f - f\|_\infty \rightarrow 0$, and multiplication by the finite measure of the common support gives convergence in L^p . For general f , choose $g \in C_c(G)$ with $\|f - g\|_p < \varepsilon$ and use the isometry of translation:

$$\|L_a f - f\|_p \leq 2\|f - g\|_p + \|L_a g - g\|_p.$$

Continuity at an arbitrary a_0 follows after translating by a_0^{-1} . $\square$

Proposition 4.3 (Approximate identity). *For $f \in L^p(G)$, $1 \leq p < \infty$,*

$$h_U * f \longrightarrow f \quad \text{in } L^p(G)$$

as U shrinks to $\{1\}$. If f is continuous, convergence is uniform on each compact set. In particular, for every compact $K \subseteq \widehat{G}$,

$$\widehat{h_U} \longrightarrow 1 \quad \text{uniformly on } K. \quad (7)$$

Proof. Minkowski's integral inequality and [theorem 4.2](#) give

$$\|h_U * f - f\|_p \leq \int_U h_U(y) \|L_y f - f\|_p d\mu_G(y),$$

which is at most ε once U is small enough. If f is continuous, the same estimate with the supremum over a fixed compact set uses uniform continuity on a compact enlargement of that set.

For (7), joint evaluation $(y, \chi) \mapsto \chi(y)$ is uniformly continuous on $C \times K$, where C is a fixed compact neighbourhood containing all sufficiently small U . Hence

$$\sup_{\chi \in K} |\widehat{h_U}(\chi) - 1| \leq \sup_{\substack{y \in U \\ \chi \in K}} |\overline{\chi(y)} - 1| \longrightarrow 0.$$

$\square$

Proposition 4.4 (The convolution Banach $*$ -algebra). *The formulas in [section 1.4](#) make $L^1(G)$ a commutative Banach $*$ -algebra, possibly without a unit. More precisely,*

$$\begin{aligned} \|f * g\|_1 &\leq \|f\|_1 \|g\|_1, & (f * g) * h &= f * (g * h), & f * g &= g * f, \\ (f * g)^* &= g^* * f^*, & \|f^*\|_1 &= \|f\|_1. \end{aligned}$$

*Moreover, if $f, g \in L^2(G)$, then $f * g$ has a continuous bounded representative and*

$$\|f * g\|_\infty \leq \|f\|_2 \|g\|_2. \quad (8)$$

Proof. Tonelli gives

$$\int_G |(f * g)(x)| dx \leq \int_G \int_G |f(y)| |g(y^{-1}x)| dy dx = \|f\|_1 \|g\|_1.$$

Fubini and the substitutions permitted by Haar invariance prove associativity, commutativity, and the involution identity. Completeness is completeness of L^1 . For $f, g \in L^2$, Cauchy-Schwarz gives (8). If $x_i \rightarrow x$, the difference is bounded by $\|L_{x_i^{-1}x} f - f\|_2 \|g\|_2$, which tends to zero by [theorem 4.2](#). $\square$

Lemma 4.5 (Positive-type kernels from convolution). *If $g \in L^2(G)$, then $g * g^*$ has a continuous bounded representative and is of positive type in the sense of [theorem 6.1](#) below. If $g \in C_c(G)$, then*

$$\text{supp}(g * g^*) \subseteq (\text{supp } g)(\text{supp } g)^{-1}.$$

Proof. Continuity and boundedness follow from [theorem 4.4](#). For $x_1, \dots, x_n \in G$ and $c_1, \dots, c_n \in \mathbb{C}$, a change of variables gives

$$\sum_{i,j} c_i \overline{c_j} (g * g^*)(x_i^{-1} x_j) = \int_G \left| \sum_i c_i g(x_i^{-1} y) \right|^2 dy \geq 0.$$

The support assertion follows directly from the convolution integral. $\square$

5 The Fourier algebra and the spectrum of $L^1(G)$

5.1 Elementary Fourier analysis

Proposition 5.1 (Fourier transform into C_0). *For $f \in L^1(G)$, the function $\widehat{f}$ of [\(1\)](#) belongs to $C_0(\widehat{G})$ and*

$$\|\widehat{f}\|_\infty \leq \|f\|_1.$$

The identities [\(4\)](#) hold.

Proof. The norm bound is immediate. If $\chi_i \rightarrow \chi$ uniformly on compact sets, choose a compact $K \subseteq G$ with $\int_{K^c} |f| < \varepsilon$. Uniform convergence on K and the bound $|\chi_i - \chi| \leq 2$ show

$$|\widehat{f}(\chi_i) - \widehat{f}(\chi)| \leq \int_K |f| |\chi_i - \chi| + 2\varepsilon \rightarrow 2\varepsilon.$$

Thus $\widehat{f}$ is continuous.

We prove vanishing at infinity first for $f \in C_c(G)$. Let $\delta = \inf\{|1 - z| : z \notin A_{\pi/4}\} > 0$. By [theorem 4.2](#), choose a relatively compact identity neighbourhood U so that $\|L_y f - f\|_1 < \delta\varepsilon$ for $y \in U$. The polar family

$$K_U = \left\{ \chi : \chi(\overline{U}) \subseteq \overline{A_{\pi/4}} \right\}$$

is compact by the Arzelà–Ascoli argument in [theorem 2.5](#). If $\chi \notin K_U$, choose $y \in \overline{U}$ with $|1 - \chi(y)| \geq \delta$. The translation identity yields

$$\delta |\widehat{f}(\chi)| \leq |1 - \overline{\chi(y)}| |\widehat{f}(\chi)| = |\widehat{L_y f - f}(\chi)| < \delta\varepsilon.$$

After making the translation estimate on $\overline{U}$ by shrinking U , this proves $|\widehat{f}| < \varepsilon$ outside a compact set. Approximation of an arbitrary $f \in L^1$ by $C_c(G)$, together with the norm bound, proves the general case.

Finally, Fubini, Haar invariance, and the substitutions $x = ay$ or $x = y^{-1}$ prove the three identities in [\(4\)](#). $\square$

Formalization note. The only delicate point in this proof is keeping $\overline{U}$ inside the region on which translation is uniformly small. One convenient Lean statement chooses a compact neighbourhood C first, proves uniform smallness on some closed $U \subseteq C$, and defines K_U using that closed U .

Let $\Delta(L^1(G))$ denote the set of nonzero complex-linear multiplicative maps $\Lambda : L^1(G) \rightarrow \mathbb{C}$, with the Gelfand topology (pointwise convergence on $L^1(G)$). Although $L^1(G)$ need not be unital, every such map is continuous: extend it to the unitization by $\widehat{\Lambda}(f, c) = \Lambda(f) + c$. If $\lambda = \widehat{\Lambda}(b)$, then $b - \lambda 1$ cannot be invertible, so $|\lambda| \leq r(b) \leq \|b\|$. Restriction gives boundedness of Λ .

Theorem 5.2 (Spectrum theorem). *The map*

$$\nu_G : \widehat{G} \longrightarrow \Delta(L^1(G)), \quad \nu_G(\chi)(f) = \widehat{f}(\chi),$$

is a homeomorphism. In particular, the complex characters of the convolution algebra $L^1(G)$ are exactly integration against conjugate group characters.

Proof. The convolution identity makes $\nu_G(\chi)$ multiplicative. It is nonzero: take a non-negative $u \in C_c(G)$ with nonzero integral and put $f = u\chi$ (using the coercion $\mathbb{T} \subset \mathbb{C}$); then $\widehat{f}(\chi) = \int u > 0$. If $\nu_G(\chi) = \nu_G(\eta)$, then $\int f(\overline{\chi} - \overline{\eta}) = 0$ for all $f \in L^1$. Testing against compactly supported functions shows $\chi = \eta$ almost everywhere, hence everywhere by continuity. Thus ν_G is injective.

Let $\Lambda \in \Delta(L^1(G))$. Choose f_0 with $\Lambda(f_0) \neq 0$ and define

$$q(a) = \frac{\Lambda(L_a f_0)}{\Lambda(f_0)}.$$

The identities

$$(L_a f) * g = L_a(f * g) = f * (L_a g)$$

and multiplicativity show, first for f or g on which Λ is nonzero and then for all of L^1 , that

$$\Lambda(L_a f) = q(a)\Lambda(f), \quad q(ab) = q(a)q(b), \quad q(1) = 1. \quad (9)$$

Continuity of translation in L^1 makes q continuous. It is bounded because $|q(a)| \leq \|\Lambda\| \|f_0\|_1 / |\Lambda(f_0)|$. A bounded homomorphism $G \rightarrow \mathbb{C}^\times$ has modulus one: for fixed a , boundedness of $q(a)^n$ and $q(a)^{-n}$ forces $|q(a)| = 1$. Hence $q : G \rightarrow \mathbb{T}$ is a character.

Bochner integration of the continuous map $a \mapsto f(a)L_a g$ in L^1 gives

$$f * g = \int_G f(a)L_a g \, da.$$

Apply Λ , use (9), and choose g with $\Lambda(g) \neq 0$:

$$\Lambda(f)\Lambda(g) = \Lambda(f * g) = \Lambda(g) \int_G f(a)q(a) \, da.$$

Thus $\Lambda(f) = \int f q = \widehat{f}(\overline{q})$, proving surjectivity.

It remains to identify the topology. Let the *transform topology* on $\widehat{G}$ be the weakest topology making all $\widehat{f}$, $f \in L^1(G)$, continuous. It is the topology transported from the Gelfand spectrum. By [theorem 5.1](#) it is no finer than the compact-open topology.

We prove joint continuity of $(a, \chi) \mapsto \chi(a)$ when $\widehat{G}$ has the transform topology. For fixed f , the map

$$(a, \chi) \longmapsto \widehat{L_a f}(\chi)$$

is continuous: near (a_0, χ_0) split the difference into a term bounded by $\|L_a f - L_{a_0} f\|_1$ and a term controlled by the defining coordinate $\widehat{L_{a_0} f}$. At (a_0, χ_0) choose f_0 with $\widehat{f_0}(\chi_0) \neq 0$. On a neighbourhood where the denominator stays nonzero,

$$\chi(a) = \frac{\widehat{L_a f_0}(\chi)}{\widehat{f_0}(\chi)}$$

is therefore continuous. Given a compact $K \subseteq G$ and an identity neighbourhood $V \subseteq \mathbb{T}$, joint continuity supplies, for every $a \in K$, a neighbourhood U_a of a and a transform-open identity neighbourhood N_a such that $\chi(U_a) \subseteq V$. A finite subcover of K shows that $\bigcap_a N_a \subseteq W(K, V)$. By [theorem 2.3](#), every compact–open identity neighbourhood is a transform-topology neighbourhood. The two topologies coincide, and ν_G is a homeomorphism. $\square$

5.2 Density and uniqueness

Put

$$\mathcal{A}(G) = \{ \widehat{f} : f \in L^1(G) \} \subseteq C_0(\widehat{G}).$$

Proposition 5.3 (Density of Fourier transforms). *$\mathcal{A}(G)$ is uniformly dense in $C_0(\widehat{G})$.*

Proof. It is a subalgebra because $\widehat{f * g} = \widehat{f} \widehat{g}$, and it is self-adjoint because $\widehat{f^*} = \overline{\widehat{f}}$. It vanishes nowhere by the construction used in [theorem 5.2](#). It separates points: if $\widehat{f}(\chi) = \widehat{f}(\eta)$ for every f , then the spectrum theorem gives $\chi = \eta$. The locally compact Stone–Weierstrass theorem (F6) proves the claim. $\square$

Theorem 5.4 (Fourier–Stieltjes uniqueness). *Let σ be a finite complex Radon measure on $\widehat{G}$. If*

$$\int_{\widehat{G}} \chi(x) d\sigma(\chi) = 0 \quad \text{for every } x \in G,$$

then $\sigma = 0$.

Proof. For $f \in L^1(G)$, Fubini gives

$$\int_{\widehat{G}} \widehat{f}(\chi) d\sigma(\chi) = \int_G f(x) \left(\int_{\widehat{G}} \overline{\chi(x)} d\sigma(\chi) \right) dx = 0,$$

because the inner integral is the assumed transform at x^{-1} . Thus σ annihilates $\mathcal{A}(G)$. By [theorem 5.3](#) it annihilates all of $C_0(\widehat{G})$, and Riesz–Markov uniqueness gives $\sigma = 0$. $\square$

Remark 5.5. This theorem is deliberately asymmetric: measures live on $\widehat{G}$, but the testing characters are only the evaluations $\chi \mapsto \chi(x)$ for $x \in G$. Surjectivity of $G \rightarrow \widehat{\widehat{G}}$ has not been used. This noncircular form is the decisive input in the final density argument for Pontryagin duality.

6 Positive-type functions and the GNS construction

Definition 6.1 (Positive type). A continuous function $\varphi : G \rightarrow \mathbb{C}$ is of *positive type* if, for every $n \geq 1$, $x_1, \dots, x_n \in G$, and $c_1, \dots, c_n \in \mathbb{C}$,

$$\sum_{i,j=1}^n \overline{c_i} c_j \varphi(x_i^{-1} x_j) \geq 0. \tag{10}$$

The inequality means that the displayed complex number is real and nonnegative. Write $P(G)$ for the convex set of positive-type functions satisfying $\varphi(1) \leq 1$.

Lemma 6.2 (Integral criterion). *For a bounded continuous φ , condition (10) is equivalent to*

$$\int_G \int_G \overline{f(x)} f(y) \varphi(x^{-1} y) d\mu_G(x) d\mu_G(y) \geq 0 \quad (f \in C_c(G)). \tag{11}$$

Proof. Assume first the finite condition. For fixed f , all variables lie in the compact set $K = \text{supp } f$. Approximate f in L^1 by compactly supported simple functions and approximate the continuous kernel $(x, y) \mapsto \varphi(x^{-1}y)$ uniformly on $K \times K$ by a step kernel obtained from a finite measurable partition. The resulting double integral is a limit of expressions

$$\sum_{i,j} \bar{d}_i d_j \varphi(t_i^{-1} t_j)$$

with the cell measures absorbed into d_i and d_j ; these are nonnegative by (10). The error is bounded by the uniform kernel error times $\|f\|_1^2$, so the limit is nonnegative.

Conversely, take a normalized bump h_U from [theorem 4.1](#) and apply (11) to $f_U = \sum_i c_i L_{x_i} h_U$. Expanding the integral gives a finite sum whose (i, j) entry is an average of φ over a set shrinking to $x_i^{-1} x_j$. Continuity of φ , compact support of h_U , and dominated convergence show that this entry tends to $\varphi(x_i^{-1} x_j)$. Passing to the limit gives (10). $\square$

Lemma 6.3 (Elementary consequences). *If φ is of positive type, then*

$$\varphi(1) \in \mathbb{R}_{\geq 0}, \quad \varphi(x^{-1}) = \overline{\varphi(x)}, \quad |\varphi(x)| \leq \varphi(1).$$

In particular, $\varphi(1) = 0$ implies $\varphi = 0$, and the definition of $P(G)$ is equivalent to the bound $\|\varphi\|_\infty \leq 1$.

Proof. The 1×1 matrix gives $\varphi(1) \geq 0$. The matrix associated to $1, x$ is positive semidefinite and therefore Hermitian, giving $\varphi(x^{-1}) = \overline{\varphi(x)}$. Its determinant is nonnegative: $\varphi(1)^2 - |\varphi(x)|^2 \geq 0$. The remaining assertions follow. $\square$

Every character $\chi \in \hat{G}$ belongs to $P(G)$, because

$$\sum_{i,j} \bar{c}_i c_j \chi(x_i^{-1} x_j) = \left| \sum_j c_j \chi(x_j) \right|^2.$$

More generally, a positive finite measure σ on $\hat{G}$ has $\check{\sigma}$ of positive type by integrating this square.

6.1 Construction of the cyclic representation

Let φ be of positive type. On $C_c(G)$ define

$$\langle f, g \rangle_\varphi = \int_G \int_G \overline{f(x)} g(y) \varphi(x^{-1} y) dx dy. \quad (12)$$

This is a positive semidefinite sesquilinear form, conjugate-linear in the first variable. Cauchy–Schwarz follows by applying nonnegativity to $f + zg$ and minimizing over $z \in \mathbb{C}$. Hence

$$N_\varphi = \{f : \langle f, f \rangle_\varphi = 0\}$$

is a linear subspace and the quotient $C_c(G)/N_\varphi$ is an inner-product space. Let H_φ be its Hilbert completion.

Proposition 6.4 (GNS representation). *There is a strongly continuous unitary representation*

$$U^\varphi : G \longrightarrow \mathcal{U}(H_\varphi)$$

and a cyclic vector $\xi_\varphi \in H_\varphi$ such that

$$\varphi(a) = \langle \xi_\varphi, U_a^\varphi \xi_\varphi \rangle \quad (a \in G). \quad (13)$$

The linear span of $\{U_a^\varphi \xi_\varphi : a \in G\}$ is dense in H_φ .

Proof. Define $U_a[f] = [L_a f]$. Haar invariance and a change of variables in (12) give $\langle L_a f, L_a g \rangle_\varphi = \langle f, g \rangle_\varphi$, so this is well-defined and extends to a unitary operator on H_φ . The translation identity gives $U_a U_b = U_{ab}$. Also

$$\| [f] \|_\varphi^2 \leq \| \varphi \|_\infty \| f \|_1^2. \quad (14)$$

Thus [theorem 4.2](#) implies $\| U_a[f] - [f] \|_\varphi \rightarrow 0$ for the dense set of classes of C_c functions. Unitarity extends this to every vector, proving strong continuity.

Define the conjugate-linear functional on $C_c(G)$

$$\Phi(f) = \int_G \overline{f(x)} \varphi(x^{-1}) dx.$$

If $f \in N_\varphi$, Cauchy-Schwarz gives $\langle f, h \rangle_\varphi = 0$ for all $h \in C_c(G)$. Put

$$F_f(y) = \int_G \overline{f(x)} \varphi(x^{-1}y) dx.$$

This is continuous and $\langle f, h \rangle_\varphi = \int h(y) F_f(y) dy$. Testing against all h gives $F_f = 0$, and in particular $\Phi(f) = F_f(1) = 0$. Thus Φ descends to the quotient.

For the approximate identity h_U ,

$$\Phi(f) = \lim_U \langle f, h_U \rangle_\varphi.$$

Indeed, the right side averages the continuous function F_f over a set shrinking to 1. Moreover, (14) and $\| h_U \|_1 = 1$ give $\| [h_U] \|_\varphi \leq \sqrt{\| \varphi \|_\infty}$. Therefore

$$|\Phi(f)| \leq \sqrt{\| \varphi \|_\infty} \| [f] \|_\varphi.$$

The functional extends continuously to H_φ . By the Hilbert-space Riesz theorem there is a unique ξ_φ such that $\Phi(f) = \langle [f], \xi_\varphi \rangle_\varphi$.

Replacing f by a translate, and using unitarity, gives for every $f \in C_c(G)$

$$\langle [f], U_a \xi_\varphi \rangle_\varphi = \int_G \overline{f(y)} \varphi(y^{-1}a) dy. \quad (15)$$

Indeed, this follows from $\langle [f], U_a \xi \rangle = \langle U_{a^{-1}}[f], \xi \rangle$ and the change of variables $y = ax$. Integrating (15) against $h(a) da$ shows, by comparison with (12), that the Bochner integral

$$\int_G h(a) U_a \xi_\varphi da$$

equals the vector $[h]$. Therefore a vector orthogonal to every $U_a \xi_\varphi$ is orthogonal to every class $[h]$, hence is zero. This proves cyclicity. Finally,

$$\langle \xi_\varphi, [h] \rangle = \overline{\Phi(h)} = \int_G h(a) \varphi(a) da,$$

where [theorem 6.3](#) was used in the last equality. Comparing this with $[h] = \int h(a) U_a \xi_\varphi da$ shows that $a \mapsto \langle \xi_\varphi, U_a \xi_\varphi \rangle$ equals $\varphi(a)$ almost everywhere. The left side is continuous by strong continuity and the right side is continuous by hypothesis, so full support of Haar measure upgrades the equality to every a . $\square$

Remark 6.5 (Measurable positive-type functions). The same proof begins with an essentially bounded measurable function satisfying the integral positivity condition. It produces the continuous representative $a \mapsto \langle \xi, U_a \xi \rangle$. This fact is needed below to make the positive unit ball weak-* closed without building continuity into a non-closed predicate.

6.2 Extreme points

Lemma 6.6 (Extreme coefficient criterion). *Let $\varphi \in P(G)$ with $\varphi(1) = 1$. Then φ is an extreme point of $P(G)$ if and only if its cyclic GNS representation is irreducible.*

Proof. Suppose first that $W \subseteq H_\varphi$ is a closed invariant subspace and let P be its orthogonal projection. Since the representation is unitary, $W^\perp$ is invariant and P commutes with every U_a . Put

$$\varphi_1(a) = \langle P\xi, U_a P\xi \rangle, \quad \varphi_2(a) = \langle (1 - P)\xi, U_a(1 - P)\xi \rangle.$$

Both are of positive type and $\varphi = \varphi_1 + \varphi_2$. If φ is extreme, the elementary extreme-point criterion for the cone slice $\{\psi : \psi(1) \leq 1\}$ says that $\varphi_1 = \lambda\varphi$ for some $0 \leq \lambda \leq 1$. Polarizing the equality of matrix coefficients after replacing a by $b^{-1}a$ gives

$$\langle U_b \xi, P U_a \xi \rangle = \lambda \langle U_b \xi, U_a \xi \rangle.$$

Cyclicity implies $P = \lambda I$. Since P is a projection, $\lambda \in \{0, 1\}$; thus W is trivial.

Conversely, if the representation is reducible, take a nontrivial invariant W . By cyclicity neither $P\xi$ nor $(1 - P)\xi$ is zero. After normalizing φ_1 and φ_2 by their positive values at 1, the equality $\varphi = \varphi_1 + \varphi_2$ becomes a nontrivial convex decomposition in $P(G)$. Hence φ is not extreme. $\square$

Lemma 6.7 (Abelian Schur lemma). *Every irreducible strongly continuous unitary representation of an abelian group on a nonzero complex Hilbert space is one-dimensional. It has the form $U_a = \chi(a)I$ for a unique $\chi \in \widehat{G}$.*

Proof. For fixed a , the unitary U_a commutes with every U_b . Every spectral projection of the normal operator U_a therefore commutes with the representation, so its range is invariant. Irreducibility forces every spectral projection to be 0 or I ; the spectral measure is concentrated at one point, and $U_a = \chi(a)I$. The representation law makes χ multiplicative, unitarity gives $|\chi| = 1$, and strong continuity gives continuity of χ . Since now every closed subspace is invariant, irreducibility forces the Hilbert space to have dimension one. $\square$

Theorem 6.8 (Extreme points of $P(G)$). *The extreme points of $P(G)$ are the zero function and the characters in $\widehat{G}$.*

Proof. The zero function is extreme because a positive-type function vanishing at 1 is zero. If a nonzero $\varphi \in P(G)$ is extreme, then necessarily $\varphi(1) = 1$; otherwise

$$\varphi = \varphi(1) \frac{\varphi}{\varphi(1)} + (1 - \varphi(1))0$$

is a nontrivial convex decomposition. By [theorems 6.6](#) and [6.7](#), its GNS representation is one-dimensional, say $U_a = \chi(a)I$. Equation [\(13\)](#) and $\|\xi\|^2 = \varphi(1) = 1$ give $\varphi = \chi$.

Conversely, the GNS space of a character is one-dimensional: in [\(12\)](#),

$$\| [f] \|_\chi^2 = \left| \int_G f(x) \chi(x) dx \right|^2.$$

It is irreducible, so the character is extreme by [theorem 6.6](#). $\square$

7 Bochner's theorem

We prove Bochner's theorem rather than use it as a disguised form of duality. Regard an essentially bounded positive-type function as an element of $L^\infty(G) = L^1(G)^*$ with its weak-* topology.

Lemma 7.1 (Weak-* compact positive ball). *The set $P(G)$, identified with its L^∞ classes, is weak-* compact and convex.*

Proof. For $f \in C_c(G)$ put

$$k_f(t) = \int_G \overline{f(x)} f(xt) dx.$$

Then $k_f \in L^1(G)$ and, by Fubini,

$$\int_G \int_G \overline{f(x)} f(y) \varphi(x^{-1}y) dx dy = \int_G \varphi(t) k_f(t) dt.$$

Thus integral positivity is the intersection, over $f \in C_c(G)$, of weak-* closed real half-spaces in the closed unit ball of L^∞ . Banach–Alaoglu makes that intersection compact. By [theorem 6.5](#), every class in it has a unique continuous positive-type representative; uniqueness uses full support. Hence it is precisely $P(G)$. Convexity is immediate. $\square$

Theorem 7.2 (Bochner). *A continuous function $\varphi : G \rightarrow \mathbb{C}$ is of positive type if and only if there is a unique finite positive Radon measure σ on $\widehat{G}$ such that*

$$\varphi(x) = \int_{\widehat{G}} \chi(x) d\sigma(\chi). \quad (16)$$

Moreover $\sigma(\widehat{G}) = \varphi(1)$.

Proof. After dividing by $\varphi(1)$, it suffices to treat $P(G)$. Let

$$\mathcal{M} = \left\{ \check{\sigma} : \sigma \geq 0, \sigma(\widehat{G}) \leq 1 \right\} \subseteq L^\infty(G).$$

The positive unit ball of finite measures on $\widehat{G} = C_0(\widehat{G})^*$ is weak-* compact. The map $\sigma \mapsto \check{\sigma}$ is continuous from that weak-* topology to the weak-* topology of $L^\infty(G)$. Indeed, for $h \in L^1(G)$, Fubini writes

$$\int_G h(x) \check{\sigma}(x) dx = \int_{\widehat{G}} \left(\int_G h(x) \chi(x) dx \right) d\sigma(\chi),$$

and the inner function of χ belongs to $C_0(\widehat{G})$ by [theorem 5.1](#). Consequently $\mathcal{M}$ is compact, hence weak-* closed. It is convex and contained in $P(G)$, since an integral of characters is of positive type.

The set $\mathcal{M}$ contains 0 and every character (use the Dirac measures). By [theorem 6.8](#), it contains every extreme point of $P(G)$. Krein–Milman says that $P(G)$ is the closed convex hull of its extreme points, so $P(G) \subseteq \mathcal{M}$. Thus equality holds. Uniqueness of σ is [theorem 5.4](#), and evaluating (16) at 1 gives its total mass. $\square$

Formalization note. The weak-* closure proof should be carried out on L^∞ equivalence classes; only after compactness is established should one choose the continuous GNS representative. This avoids attempting to prove that continuity is weak-* closed. The use of the spectral theorem in [theorem 6.7](#) is generic Hilbert-space functional analysis and may be packaged as a separate Schur lemma.

8 Construction of the dual Haar measure and Fourier inversion

Let

$$V(G) = \text{span}_{\mathbb{C}} P(G), \quad V^1(G) = V(G) \cap L^1(G).$$

By linearity of Bochner's theorem, for every $f \in V(G)$ there is a finite complex measure σ_f on $\widehat{G}$ such that

$$f(x) = \int_{\widehat{G}} \chi(x) d\sigma_f(\chi). \quad (17)$$

To see that this measure is independent of a presentation of f as a linear combination of positive-type functions, subtract two candidate measures and apply [theorem 5.4](#). Thus $f \mapsto \sigma_f$ is a well-defined linear map. When f is of positive type, σ_f is positive.

8.1 A symmetric measure identity

Lemma 8.1. *For $f, g \in V^1(G)$,*

$$\widehat{g} \sigma_f = \widehat{f} \sigma_g \quad \text{as finite complex measures on } \widehat{G}. \quad (18)$$

Proof. Both products are finite measures because Fourier transforms are bounded. It suffices by [theorem 5.4](#) to compare their inverse Fourier–Stieltjes transforms. Fubini and (17) give

$$\begin{aligned} \int_{\widehat{G}} \chi(x) \widehat{g}(\chi) d\sigma_f(\chi) &= \int_G g(y) \left(\int_{\widehat{G}} \chi(xy^{-1}) d\sigma_f(\chi) \right) dy \\ &= \int_G g(y) f(xy^{-1}) dy = (g * f)(x). \end{aligned}$$

The other measure has transform $(f * g)(x)$, equal to this because G is abelian. $\square$

Choose the approximate identity h_U of [theorem 4.3](#) and put

$$e_U = h_U * h_U^*. \quad (19)$$

By [theorem 4.5](#), $e_U \in V^1(G)$ is of positive type, so σ_{e_U} is positive. Moreover

$$\widehat{e_U} = |\widehat{h_U}|^2 \longrightarrow 1 \quad \text{uniformly on compact subsets of } \widehat{G}. \quad (20)$$

8.2 The auxiliary multiplier class

Define $\mathcal{F}$ to be the set of $u \in C_b(\widehat{G})$ for which there exists a finite complex measure τ_u on $\widehat{G}$ satisfying

$$u \sigma_f = \widehat{f} \tau_u \quad \text{for every } f \in V^1(G). \quad (21)$$

Products of a function and a measure mean multiplication of the measure by that bounded Borel function.

Lemma 8.2 (Multiplier calculus). *The class $\mathcal{F}$ has the following properties.*

(i) *The measure τ_u is unique. If $u \geq 0$, then $\tau_u \geq 0$.*

(ii) *$\mathcal{F}$ is a $C_b(\widehat{G})$ -module, and*

$$\tau_{u+v} = \tau_u + \tau_v, \quad \tau_{cu} = c\tau_u, \quad \tau_{au} = a\tau_u$$

whenever the expressions make sense.

(iii) For $\eta \in \widehat{G}$, set $(T_\eta u)(\chi) = u(\eta^{-1}\chi)$ and let $T_\eta \tau$ be pushforward under $\chi \mapsto \eta\chi$. Then $T_\eta u \in \mathcal{F}$ and $\tau_{T_\eta u} = T_\eta \tau_u$.

(iv) If $g \in V^1(G)$, then $\widehat{g} \in \mathcal{F}$ and $\tau_{\widehat{g}} = \sigma_g$.

(v) $C_c(\widehat{G}) \subseteq \mathcal{F}$.

Proof. Apply (21) with $f = e_U$. For $a \in C_c(\widehat{G})$,

$$\int a d\tau_u = \lim_U \int a \widehat{e_U} d\tau_u = \lim_U \int a u d\sigma_{e_U}. \quad (22)$$

The first limit follows from uniform convergence on $\text{supp } a$. The last expression is determined by u , proving uniqueness. If $u \geq 0$ and $a \geq 0$, it is nonnegative because σ_{e_U} is positive; Riesz–Markov then gives $\tau_u \geq 0$.

Linearity follows immediately from (21) and uniqueness. If $a \in C_b(\widehat{G})$, multiply both sides of (21) by a to obtain $(au)\sigma_f = \widehat{f}(a\tau_u)$; this proves the module assertion.

For translation, define the modulation $(M_\eta f)(x) = \eta(x)f(x)$. Directly from (17) and (1),

$$\sigma_{M_\eta f} = T_\eta \sigma_f, \quad \widehat{M_\eta f} = T_\eta \widehat{f}. \quad (23)$$

Apply (21) to $M_{\eta^{-1}} f$ and push the resulting equality forward by T_η . This gives $(T_\eta u)\sigma_f = \widehat{f}(T_\eta \tau_u)$.

Assertion (iv) is exactly [theorem 8.1](#): for all f , $\widehat{g}\sigma_f = \widehat{f}\sigma_g$.

Finally let $\gamma \in C_c(\widehat{G})$ and $K = \text{supp } \gamma$. By (20), choose U with $|\widehat{e_U} - 1| < 1/2$ on K . Hence $\widehat{e_U}$ is nonzero on an open neighbourhood of K . The function

$$a(\chi) = \begin{cases} \gamma(\chi)/\widehat{e_U}(\chi), & \widehat{e_U}(\chi) \neq 0, \\ 0, & \widehat{e_U}(\chi) = 0 \end{cases}$$

is continuous and compactly supported: near the zero set of $\widehat{e_U}$ it is zero because that set is disjoint from $K = \text{supp } \gamma$. Thus $a \in C_b(\widehat{G})$ and $\gamma = a\widehat{e_U}$. Assertions (ii) and (iv) imply $\gamma \in \mathcal{F}$. $\square$

8.3 The dual measure

For $\gamma \in C_c(\widehat{G})$ define

$$\Lambda(\gamma) = \tau_\gamma(\widehat{G}). \quad (24)$$

This is well-defined by [theorem 8.2](#).

Proposition 8.3 (Dual Haar measure). *Λ is a nonzero positive translation-invariant linear functional on $C_c(\widehat{G})$. Therefore there is a nonzero Haar measure $\mu_{\widehat{G}}$ such that*

$$\Lambda(\gamma) = \int_{\widehat{G}} \gamma(\chi) d\mu_{\widehat{G}}(\chi) \quad (\gamma \in C_c(\widehat{G})). \quad (25)$$

Proof. Linearity and positivity follow from parts (i)–(ii) of [theorem 8.2](#). A positive functional on C_c is locally bounded: if K is compact, choose $b \in C_c$, $b \geq 1$ on K ; decomposing a complex γ supported in K into four nonnegative parts bounds $|\Lambda(\gamma)|$ by a fixed multiple of $\|\gamma\|_\infty \Lambda(b)$. Riesz–Markov therefore supplies a positive Radon measure satisfying (25).

It is nonzero. Choose nonzero $g \in C_c(G)$ and put $f = g * g^*$. Then f is nonzero, integrable, and of positive type, so σ_f is a nonzero positive measure. Choose $a \in C_c(\widehat{G})$, $a \geq 0$, with $\int a d\sigma_f > 0$. Since $\widehat{f} \in \mathcal{F}$ and $a\widehat{f} \in C_c(\widehat{G})$,

$$\Lambda(a\widehat{f}) = \tau_{a\widehat{f}}(\widehat{G}) = \int a d\sigma_f > 0.$$

For $\eta \in \widehat{G}$, part (iii) of [theorem 8.2](#) gives $\Lambda(T_\eta \gamma) = (T_\eta \tau_\gamma)(\widehat{G}) = \Lambda(\gamma)$. Thus the representing measure is translation invariant and nonzero, hence is Haar. $\square$

Theorem 8.4 (Fourier inversion on V^1). *For every $f \in V^1(G)$,*

$$\sigma_f = \widehat{f} \mu_{\widehat{G}}, \quad \widehat{f} \in L^1(\widehat{G}), \quad f(x) = \int_{\widehat{G}} \widehat{f}(\chi) \chi(x) d\mu_{\widehat{G}}(\chi). \quad (26)$$

The Haar measure $\mu_{\widehat{G}}$ is uniquely determined by this property.

Proof. We first prove, for every $u \in \mathcal{F}$, that

$$\tau_u = u \mu_{\widehat{G}}. \quad (27)$$

If $a \in C_c(\widehat{G})$, then $au \in C_c(\widehat{G})$ and the module identity gives

$$\int a(\chi) u(\chi) d\mu_{\widehat{G}}(\chi) = \Lambda(au) = \tau_{au}(\widehat{G}) = \int a d\tau_u.$$

Riesz–Markov uniqueness proves (27). Apply it to $u = \widehat{f}$, for which $\tau_u = \sigma_f$ by [theorem 8.2](#). Since σ_f is finite, $\widehat{f} \in L^1(\widehat{G})$, and substituting its density in (17) proves inversion.

If ν is another Haar measure with the same inversion property, then $\widehat{f} \nu = \sigma_f = \widehat{f} \mu_{\widehat{G}}$ for every $f \in V^1$. Given $a \in C_c(\widehat{G})$, choose e_U so that $\widehat{e_U}$ has no zero on $\text{supp } a$. Testing the measure equality for e_U against $a/\widehat{e_U}$ gives $\int a d\nu = \int a d\mu_{\widehat{G}}$. Hence ν equals $\mu_{\widehat{G}}$. $\square$

Corollary 8.5 (Positivity and scaling). *If $f \in V^1(G)$ is of positive type, then $\widehat{f} \geq 0$ everywhere. If the original Haar measure is replaced by $c\mu_G$ with $c > 0$, the dual Haar measure selected by [theorem 8.4](#) is replaced by $c^{-1}\mu_{\widehat{G}}$.*

Proof. For positive-type f , the measure $\sigma_f = \widehat{f} \mu_{\widehat{G}}$ is positive, so $\widehat{f} \geq 0$ almost everywhere. It is continuous and Haar measure has full support, so the inequality is pointwise. Under scaling by c , the Fourier transform is multiplied by c ; multiplying the dual measure by c^{-1} leaves the inversion integral unchanged. Uniqueness gives the assertion. $\square$

9 Plancherel theory and localized Fourier transforms

The only purpose of the full L^2 theorem in the biduality proof is to manufacture a nonzero Fourier transform supported in a prescribed nonempty open set. We nevertheless record the complete extension because it is a central part of the theory and fixes the normalization of dual Haar measure.

Theorem 9.1 (Parseval on the dense domain). *If $f \in L^1(G) \cap L^2(G)$, then $\widehat{f} \in L^2(\widehat{G})$ and*

$$\int_G |f(x)|^2 d\mu_G(x) = \int_{\widehat{G}} |\widehat{f}(\chi)|^2 d\mu_{\widehat{G}}(\chi). \quad (28)$$

Proof. Set $k = f * f^*$. By [theorem 4.4](#), $k \in L^1(G)$ is continuous; by [theorem 4.5](#), it is of positive type. Hence $k \in V^1(G)$. Furthermore

$$k(1) = \|f\|_2^2, \quad \widehat{k} = \widehat{f} \widehat{f}^* = |\widehat{f}|^2.$$

Apply inversion ([26](#)) to k at 1. □

Lemma 9.2 (Density of the Fourier range). *The image of $L^1(G) \cap L^2(G)$ under the Fourier transform is dense in $L^2(\widehat{G})$.*

Proof. Suppose $H \in L^2(\widehat{G})$ is orthogonal to the image. Fix $f \in L^1(G) \cap L^2(G)$. For every $x \in G$, the function $\chi \mapsto \chi(x)\widehat{f}(\chi)$ is the Fourier transform of $L_{x^{-1}}f$ and is again in the image. Therefore

$$\int_{\widehat{G}} \overline{H(\chi)} \widehat{f}(\chi) \chi(x) d\mu_{\widehat{G}}(\chi) = 0 \quad (x \in G).$$

The density $\overline{H}\widehat{f}$ belongs to $L^1(\widehat{G})$ by Cauchy–Schwarz, so it defines a finite measure. Fourier–Stieltjes uniqueness [theorem 5.4](#) gives

$$\overline{H}\widehat{f} = 0 \quad \text{almost everywhere} \quad \text{for every } f \in L^1 \cap L^2. \quad (29)$$

If H were nonzero, regularity of the finite measure $|H|^2 \mu_{\widehat{G}}$ would give a compact set K on which $\int_K |H|^2 > 0$. Choose e_U from ([19](#)) so that $|\widehat{e_U} - 1| < 1/2$ on K . Since $e_U \in C_c(G) \subseteq L^1 \cap L^2$, equation (29) applied to e_U contradicts the choice of K . Thus $H = 0$, and the orthogonal complement of the image is trivial. □

Theorem 9.3 (Plancherel). *The Fourier transform extends uniquely to a unitary operator*

$$\mathcal{P}_G : L^2(G, \mu_G) \longrightarrow L^2(\widehat{G}, \mu_{\widehat{G}}).$$

For $f, g \in L^2(G)$,

$$\int_G \overline{f(x)} g(x) dx = \int_{\widehat{G}} \overline{\mathcal{P}_G f(\chi)} \mathcal{P}_G g(\chi) d\chi. \quad (30)$$

Proof. The space $L^1 \cap L^2$ is dense in L^2 , for instance because $C_c(G)$ is dense. By [theorem 9.1](#), Fourier transform is an isometry there, so it extends uniquely to an isometry on L^2 . Its range is closed and is dense by [theorem 9.2](#), hence is all of $L^2(\widehat{G})$. Polarization gives (30). □

Lemma 9.4 (L^2 product–convolution formula). *If $a, b \in L^2(G)$, then $ab \in L^1(G)$ and*

$$\widehat{ab} = (\mathcal{P}_G a) * (\mathcal{P}_G b), \quad (31)$$

where the convolution on the right is the continuous bounded representative of the convolution of two $L^2(\widehat{G})$ functions.

Proof. Hölder gives $ab \in L^1$. Fix $\eta \in \widehat{G}$ and put $d(x) = \overline{b(x)}\eta(x)$. Then $d \in L^2(G)$ and a direct calculation on the dense domain $L^1 \cap L^2$, followed by L^2 approximation, gives

$$(\mathcal{P}_G b)(\chi^{-1}\eta) = \overline{(\mathcal{P}_G d)(\chi)}.$$

Consequently Parseval yields

$$\begin{aligned} ((\mathcal{P}_G a) * (\mathcal{P}_G b))(\eta) &= \int_{\widehat{G}} (\mathcal{P}_G a)(\chi) (\mathcal{P}_G b)(\chi^{-1}\eta) d\chi \\ &= \int_{\widehat{G}} (\mathcal{P}_G a)(\chi) \overline{(\mathcal{P}_G d)(\chi)} d\chi \\ &= \int_G a(x) \overline{d(x)} dx = \int_G a(x) b(x) \overline{\eta(x)} dx = \widehat{ab}(\eta). \end{aligned}$$

Both sides have canonical continuous representatives, so the pointwise identity holds everywhere. $\square$

Proposition 9.5 (Fourier transform localized in an open set). *Let U be a nonempty open subset of $\widehat{G}$. There exists $F \in L^1(G)$ such that $\widehat{F}$ is nonzero and*

$$\text{supp}(\widehat{F}) \subseteq U.$$

Proof. Haar measure is positive on U and inner regular, so choose compact $K \subseteq U$ with $0 < \mu_{\widehat{G}}(K) < \infty$. Compactness of K and continuity of multiplication give an identity neighbourhood W with $WK \subseteq U$. Choose a compact neighbourhood B of 1 contained in W ; then $0 < \mu_{\widehat{G}}(B) < \infty$. The indicator functions $\mathbf{1}_B, \mathbf{1}_K$ belong to $L^2(\widehat{G})$. Their convolution

$$q = \mathbf{1}_B * \mathbf{1}_K$$

has a continuous representative, is supported in the compact set $BK \subseteq U$, and is nonzero because

$$\int_{\widehat{G}} q = \mu_{\widehat{G}}(B)\mu_{\widehat{G}}(K) > 0.$$

By Plancherel choose $a, b \in L^2(G)$ whose transforms are $\mathbf{1}_B$ and $\mathbf{1}_K$. Then $F = ab \in L^1(G)$ and [theorem 9.4](#) gives $\widehat{F} = q$. $\square$

Formalization note. For indicator functions it is safest to choose compact measurable B, K with positive finite measure and prove that the continuous convolution representative vanishes off BK . The theorem is used once with the LCA group $\widehat{G}$ and an open subset of $\widehat{G}$.

10 Pontryagin biduality

We now prove that the continuous homomorphism α_G of [theorem 3.1](#) is a topological isomorphism. The proof is divided into four properties so that each can become an independent Lean theorem.

10.1 Characters separate points

Proposition 10.1 (Injectivity). $\alpha_G : G \rightarrow \widehat{\widehat{G}}$ is injective.

Proof. Suppose $z \in G$ satisfies $\chi(z) = 1$ for every $\chi \in \widehat{G}$. If $z \neq 1$, choose an identity neighbourhood U with $zU \cap U = \emptyset$. Choose a relatively compact identity neighbourhood V with $VV^{-1} \subseteq U$, and a nonzero $g \in C_c(G)$ supported in V . The function

$$f = \frac{g * g^*}{(g * g^*)(1)}$$

is nonzero, belongs to $V^1(G)$, is of positive type, satisfies $f(1) = 1$, and has support in U . Fourier inversion gives

$$f(z^{-1}x) = \int_{\widehat{G}} \widehat{f}(\chi) \chi(z)^{-1} \chi(x) d\chi = f(x)$$

for every x . Thus $L_z f = f$. But $\text{supp}(L_z f) = z \text{supp } f \subseteq zU$ is disjoint from $\text{supp } f \subseteq U$, impossible for a nonzero function. Hence $z = 1$. $\square$

Corollary 10.2. For $x \neq y$ in G , there is $\chi \in \widehat{G}$ with $\chi(x) \neq \chi(y)$.

Proof. Apply [theorem 10.1](#) to $x^{-1}y \neq 1$. $\square$

10.2 The original topology is recovered from characters

For compact $K \subseteq \widehat{G}$ and an identity neighbourhood $V \subseteq \mathbb{T}$, put

$$K_V^\triangleleft = \{x \in G : \chi(x) \in V \text{ for every } \chi \in K\} = \alpha_G^{-1}(W(K, V)).$$

Proposition 10.3 (Topological embedding). *The sets $K_V^\triangleleft$ form a neighbourhood basis at 1 in G . Consequently α_G is a homeomorphism from G onto its image in $\widehat{\widehat{G}}$.*

Proof. Each $K_V^\triangleleft$ is a neighbourhood of 1 because α_G is continuous. We prove cofinality in the other direction. Let U be an identity neighbourhood. As in [theorem 10.1](#), choose a continuous positive-type $f \in V^1(G)$ with

$$f(1) = 1, \quad \text{supp } f \subseteq U.$$

By [theorems 8.4](#) and [8.5](#), $\widehat{f} \geq 0$ and

$$1 = f(1) = \int_{\widehat{G}} \widehat{f}(\chi) d\chi.$$

Choose $0 < \varepsilon < 1/3$. Regularity of the finite measure $\widehat{f} \mu_{\widehat{G}}$ gives a compact $K \subseteq \widehat{G}$ such that

$$\int_{K^c} \widehat{f}(\chi) d\chi < \varepsilon. \quad (32)$$

Let $V = \{z \in \mathbb{T} : |z - 1| < \varepsilon\}$. If $x \in K_V^\triangleleft$, inversion and (32) give

$$\begin{aligned} |f(x) - 1| &\leq \int_K \widehat{f}(\chi) |\chi(x) - 1| d\chi + \int_{K^c} \widehat{f}(\chi) |\chi(x) - 1| d\chi \\ &< \varepsilon \int_K \widehat{f} + 2\varepsilon < 3\varepsilon < 1. \end{aligned}$$

Thus $f(x) \neq 0$, so $x \in \text{supp } f \subseteq U$. We have found $K_V^\triangleleft \subseteq U$, proving the basis assertion. Since the subspace neighbourhoods of 1 in $\alpha_G(G)$ are precisely the intersections with $W(K, V)$, the inverse of α_G on its image is continuous. $\square$

10.3 The image is closed

Lemma 10.4 (Dense locally compact subspaces). *If X is a dense locally compact subspace of a Hausdorff space Y , then X is open in Y .*

Proof. Fix $x \in X$. Choose a compact neighbourhood K of x in X and a relatively open set $V \subseteq X$ with $x \in V \subseteq K$. Write $V = X \cap O$ for an open $O \subseteq Y$. The set K is compact in Y , hence closed. If $O \setminus K$ were nonempty, it would be a nonempty open subset of Y and would meet the dense set X . But

$$X \cap (O \setminus K) = V \setminus K = \emptyset,$$

a contradiction. Hence $x \in O \subseteq K \subseteq X$, so X is open. $\square$

Proposition 10.5 (Closed image). *The subgroup $\alpha_G(G)$ is closed in $\widehat{\widehat{G}}$.*

Proof. Let $H = \overline{\alpha_G(G)}$. By [theorem 10.3](#), $\alpha_G(G)$ is locally compact in its subspace topology; it is dense in the Hausdorff space H . By [theorem 10.4](#), it is open in H . An open subgroup is closed, since its complement is the union of the other open cosets. Thus $\alpha_G(G)$ is both dense and closed in H , so it equals H and is closed in $\widehat{\widehat{G}}$. $\square$

10.4 The image is all of the bidual

Proposition 10.6 (Surjectivity). $\alpha_G : G \rightarrow \widehat{\widehat{G}}$ is surjective.

Proof. Suppose not. By [theorem 10.5](#),

$$\Omega = \widehat{\widehat{G}} \setminus \alpha_G(G)$$

is a nonempty open set. Apply [theorem 9.5](#) to the LCA group $\widehat{\widehat{G}}$. There is $\Phi \in L^1(\widehat{\widehat{G}})$ whose Fourier transform on $\widehat{\widehat{G}}$ is nonzero and supported in Ω . It therefore vanishes at every evaluation character:

$$0 = \widehat{\Phi}(\alpha_G(x)) = \int_{\widehat{\widehat{G}}} \Phi(\chi) \overline{\alpha_G(x)(\chi)} d\chi = \int_{\widehat{\widehat{G}}} \Phi(\chi) \chi(x^{-1}) d\chi. \quad (33)$$

The finite complex measure $\Phi \mu_{\widehat{\widehat{G}}}$ therefore has zero inverse Fourier–Stieltjes transform at every $x \in G$. The asymmetric uniqueness theorem [theorem 5.4](#)—which does not assume biduality—implies $\Phi \mu_{\widehat{\widehat{G}}} = 0$. Hence $\Phi = 0$ almost everywhere and $\widehat{\Phi} = 0$, contradicting its construction. Thus Ω is empty. $\square$

Theorem 10.7 (Pontryagin duality). *For every LCA group G , evaluation is a natural isomorphism of topological groups*

$$\alpha_G : G \xrightarrow{\sim} \widehat{\widehat{G}}, \quad \alpha_G(x)(\chi) = \chi(x). \quad (34)$$

It is natural in G , and the triangle identity $(\widehat{\alpha_G})\alpha_{\widehat{G}} = \text{id}_{\widehat{G}}$ holds.

Proof. Continuity is [theorem 3.1](#); injectivity, continuity of the inverse, and surjectivity are [theorems 10.1](#), [10.3](#) and [10.6](#). Naturality and the triangle identity are [theorem 3.2](#). $\square$

Corollary 10.8 (Mutual normalization of Haar measures). *Start with μ_G , construct $\mu_{\widehat{G}}$ by [theorem 8.3](#), and then construct the dual of $\mu_{\widehat{G}}$. Under $\alpha_G : G \simeq \widehat{\widehat{G}}$, the resulting Haar measure on $\widehat{\widehat{G}}$ is exactly μ_G , not merely a scalar multiple.*

Proof. Transport the second dual measure along α_G ; it is $c\mu_G$ for some $c > 0$ by Haar uniqueness. For $F \in L^1(\widehat{G}) \cap L^2(\widehat{G})$, the adjoint formula for the unitary Plancherel operator gives

$$\widehat{F}(\alpha_G(x)) = (\mathcal{P}_G^{-1}F)(x^{-1}).$$

Apply Plancherel for the group $\widehat{G}$ and then change variables by inversion:

$$\|F\|_{2,\mu_{\widehat{G}}}^2 = \|\widehat{F}\|_{2,\mu_{\widehat{\widehat{G}}}}^2 = c\|\mathcal{P}_G^{-1}F\|_{2,\mu_G}^2 = c\|F\|_{2,\mu_{\widehat{G}}}^2.$$

Choose $F \neq 0$ to obtain $c = 1$. $\square$

11 Annihilators, subgroups, quotients, and exactness

For $S \subseteq G$ and $T \subseteq \widehat{G}$ define

$$S^\perp = \{\chi \in \widehat{G} : \chi(s) = 1 \text{ for all } s \in S\},$$

$$T^\perp = \{x \in G : \chi(x) = 1 \text{ for all } \chi \in T\}.$$

The second definition uses the evaluation embedding, now known to identify G with its bidual.

Lemma 11.1 (Elementary annihilator laws). *For arbitrary subsets $S, S_i \subseteq G$:*

- (a) $S^\perp$ is a closed subgroup of $\widehat{G}$;
- (b) inclusion reverses: $S \subseteq T$ implies $T^\perp \subseteq S^\perp$;
- (c) $S^\perp = (\overline{\langle S \rangle})^\perp$, where $\overline{\langle S \rangle}$ is the closed subgroup generated by S ;
- (d) $(\bigcup_i S_i)^\perp = \bigcap_i S_i^\perp$.

The analogous statements hold for subsets of $\widehat{G}$.

Proof. The annihilator is the intersection over $s \in S$ of the kernels of the continuous evaluation homomorphisms $\chi \mapsto \chi(s)$, hence is a closed subgroup. The other claims follow from the definitions, continuity of characters, and the fact that a character trivial on S is trivial on the subgroup generated by S and then on its closure. $\square$

Let H be a closed subgroup of G , and let $q : G \rightarrow G/H$ be the quotient map. The quotient is again an LCA group.

Theorem 11.2 (Dual of a quotient). *Precomposition with q is a topological isomorphism*

$$\widehat{q} : (\widehat{G/H}) \xrightarrow{\sim} H^\perp. \quad (35)$$

Proof. A character ψ of G/H pulls back to a character of G trivial on H . Conversely, if $\chi \in H^\perp$, then $\bar{\chi}(gH) = \chi(g)$ is well-defined and is a homomorphism. It is continuous because q is a quotient map and $\bar{\chi} \circ q = \chi$. These constructions are inverse, proving algebraic bijectivity. Continuity of $\widehat{q}$ is functoriality of the compact–open topology.

For continuity of the inverse we use the following compact lifting fact. Every compact $C \subseteq G/H$ is contained in $q(K)$ for some compact $K \subseteq G$. Indeed, choose a compact identity neighbourhood $B \subseteq G$. The open sets $yq(B^\circ)$ cover C ; choose finitely many, lift their translating elements to $x_1, \dots, x_n \in G$, and take $K = \bigcup_i x_i B$.

Now a compact–open neighbourhood $W(C, V)$ in $(\widehat{G/H})$ contains the inverse image of $W(K, V) \cap H^\perp$ under (35), because $C \subseteq q(K)$. Thus the inverse is continuous at the identity and hence everywhere. $\square$

Theorem 11.3 (Double annihilator). *If H is a closed subgroup of G , then*

$$(H^\perp)^\perp = H. \quad (36)$$

For an arbitrary subset $S \subseteq G$,

$$(S^\perp)^\perp = \overline{\langle S \rangle}. \quad (37)$$

Proof. The inclusion $H \subseteq (H^\perp)^\perp$ is immediate. If $x \notin H$, then $q(x) \neq 1$ in the LCA group G/H . By theorem 10.2, choose $\psi \in (\widehat{G/H})$ with $\psi(q(x)) \neq 1$. Its pullback $\psi \circ q$ belongs to $H^\perp$ and does not vanish at x . Thus $x \notin (H^\perp)^\perp$, proving (36). The general formula follows from theorem 11.1 by replacing S with $\langle S \rangle$. $\square$

Corollary 11.4 (Lattice formulas). *For closed subgroups $H, K \leq G$,*

$$(\overline{HK})^\perp = H^\perp \cap K^\perp, \quad (38)$$

$$(H \cap K)^\perp = \overline{H^\perp K^\perp}. \quad (39)$$

The same formulas hold for arbitrary families, with closed generated subgroup replacing the finite product.

Proof. The first identity is immediate from the definition. Let $A = \overline{H^\perp K^\perp}$. Applying the first identity in the dual group and then double annihilation gives

$$A^\perp = (H^\perp)^\perp \cap (K^\perp)^\perp = H \cap K.$$

Taking annihilators once more gives (39). The family version is identical. $\square$

11.1 Extension of characters

We first isolate a useful formal consequence of biduality.

Lemma 11.5 (The dual reflects isomorphisms). *If $u : A \rightarrow B$ is a continuous homomorphism of LCA groups and $\hat{u} : \hat{B} \rightarrow \hat{A}$ is a topological isomorphism, then u is a topological isomorphism.*

Proof. Naturality (5) rearranges to

$$u = \alpha_B^{-1} \circ \hat{u} \circ \alpha_A. \quad (40)$$

If $\hat{u}$ is an isomorphism, so is its dual $\hat{\hat{u}}$; the two evaluation maps are isomorphisms by theorem 10.7. Hence the right side is an isomorphism. $\square$

Theorem 11.6 (Character extension and dual of a subgroup). *Let $i : H \hookrightarrow G$ be the inclusion of a closed subgroup. Restriction*

$$\hat{i} : \hat{G} \longrightarrow \hat{H}, \quad \chi \longmapsto \chi|_H$$

is a continuous open surjection with kernel $H^\perp$. Equivalently, every continuous character of H extends to a continuous character of G , and restriction induces a topological isomorphism

$$\hat{G}/H^\perp \xrightarrow{\sim} \hat{H}. \quad (41)$$

Proof. The kernel is visibly $H^\perp$. Put $N = H^\perp$, $Q = \hat{G}/N$, and let $\bar{r} : Q \rightarrow \hat{H}$ be the injective map induced by restriction. It is continuous by the quotient property. We prove that its dual is an isomorphism.

Apply theorem 11.2 to the closed subgroup $N \leq \hat{G}$. It identifies $\hat{Q}$ with $N^\perp \subseteq \hat{\hat{G}}$. By theorem 11.3,

$$N^\perp = (H^\perp)^\perp = \alpha_G(H).$$

Consequently the map

$$\theta : H \longrightarrow \hat{Q}, \quad \theta(h)(\chi N) = \chi(h)$$

is a topological isomorphism. For $h \in H$ and $\chi N \in Q$,

$$\begin{aligned} ((\hat{r})(\alpha_H(h)))(\chi N) &= \alpha_H(h)(\bar{r}(\chi N)) \\ &= (\chi|_H)(h) = \chi(h) = \theta(h)(\chi N). \end{aligned}$$

Thus $(\hat{r}) \circ \alpha_H = \theta$ is an isomorphism. Since α_H is an isomorphism, $\hat{r}$ is one too. By theorem 11.5, $\bar{r}$ is a topological isomorphism. In particular restriction is surjective and, as the composite of the open quotient map $\hat{G} \rightarrow Q$ with an isomorphism, is open. $\square$

Theorem 11.7 (Exactness). *For a closed subgroup $H \leq G$, the strict exact sequence*

$$1 \longrightarrow H \xrightarrow{i} G \xrightarrow{q} G/H \longrightarrow 1$$

dualizes to the strict exact sequence

$$1 \longrightarrow (\widehat{G/H}) \xrightarrow{\widehat{q}} \widehat{G} \xrightarrow{\widehat{i}} \widehat{H} \longrightarrow 1. \quad (42)$$

Here $\widehat{q}$ is a closed embedding and $\widehat{i}$ is an open quotient map.

Proof. By [theorem 11.2](#), $\widehat{q}$ identifies the first group with $H^\perp$. By [theorem 11.6](#), this subgroup is the kernel of the surjective open map $\widehat{i}$. These statements give algebraic exactness and the asserted topology. $\square$

Proposition 11.8 (Kernel and closure of the dual image). *For any continuous homomorphism $f : G \rightarrow H$,*

$$\ker(\widehat{f}) = (\overline{f(G)})^\perp, \quad (43)$$

$$\overline{(\widehat{f})(\widehat{H})} = (\ker f)^\perp. \quad (44)$$

If f occurs as the inclusion or quotient map in a strict exact sequence, the closure in (44) may be omitted.

Proof. A character $\psi \in \widehat{H}$ lies in the kernel precisely when it is trivial on $f(G)$, and continuity makes this equivalent to triviality on its closure. For the second formula, $x \in G$ annihilates the image of $\widehat{f}$ exactly when

$$\psi(f(x)) = 1 \quad \text{for every } \psi \in \widehat{H}.$$

Characters separate points of H , so this is equivalent to $f(x) = 1$. Thus the annihilator of the image is $\ker f$; take annihilators again and use [theorem 11.3](#). The last assertion follows from [theorems 11.2](#) and [11.6](#). $\square$

12 Categorical and structural consequences

Theorem 12.1 (Contravariant equivalence). *Pontryagin duality is a contravariant equivalence on the category of LCA groups and continuous homomorphisms. Its object map is $G \mapsto \widehat{G}$, its morphism map is $f \mapsto \widehat{f}$, and its unit is the natural isomorphism $\alpha : \text{id} \Rightarrow (\widehat{})$.*

Proof. Contravariant identity and composition laws were proved in [section 3](#). The maps α_G are isomorphisms by [theorem 10.7](#), are natural by (5), and obey the triangle identity (6). These data are precisely a dual equivalence.

Explicitly, the functor is faithful: if $\widehat{f} = \widehat{g}$, then for every $x \in G$ and $\psi \in \widehat{H}$, $\psi(f(x)) = (\widehat{f}\psi)(x) = (\widehat{g}\psi)(x) = \psi(g(x))$; characters separate points, so $f = g$. It is full in the contravariant sense. Given a continuous $u : \widehat{H} \rightarrow \widehat{G}$, define

$$f = \alpha_H^{-1} \circ \widehat{u} \circ \alpha_G : G \longrightarrow H. \quad (45)$$

For $x \in G$ and $\psi \in \widehat{H}$,

$$\psi(f(x)) = (\widehat{u}(\alpha_G(x)))(\psi) = \alpha_G(x)(u(\psi)) = u(\psi)(x),$$

so $\widehat{f} = u$ by extensionality. $\square$

Theorem 12.2 (Compact/discrete equivalence). *For an LCA group G ,*

$$\begin{aligned} G \text{ is compact} &\iff \widehat{G} \text{ is discrete,} \\ G \text{ is discrete} &\iff \widehat{G} \text{ is compact.} \end{aligned}$$

Thus Pontryagin duality restricts to a dual equivalence between compact abelian groups and discrete abelian groups.

Proof. The forward implications are [theorem 2.6](#). If $\widehat{G}$ is discrete, then $\widehat{\widehat{G}}$ is compact, and $G \simeq \widehat{\widehat{G}}$ is compact. If $\widehat{G}$ is compact, then $\widehat{\widehat{G}}$ is discrete, and hence so is G . $\square$

Corollary 12.3. *An LCA group is both compact and discrete if and only if it is finite. In that case its dual is finite of the same cardinality.*

Proof. A compact discrete space has a finite cover by singleton open sets. Conversely a finite Hausdorff group is compact and discrete. For finite G , biduality gives $|G| = |\widehat{\widehat{G}}|$; finite character theory, or the cyclic decomposition in [theorem 13.5](#), gives $|\widehat{G}| = |G|$. $\square$

Proposition 12.4 (Closed subgroup tests). *For a closed subgroup $H \leq G$:*

- (a) *if H is open, then $H^\perp$ is compact;*
- (b) *if H is compact, then $H^\perp$ is open in $\widehat{G}$;*
- (c) *G/H is compact if and only if $H^\perp$ is discrete;*
- (d) *G/H is discrete if and only if $H^\perp$ is compact.*

Proof. If H is open, G/H is discrete, so $H^\perp \simeq \widehat{(G/H)}$ is compact. If H is compact, $\widehat{H}$ is discrete and restriction $\widehat{G} \rightarrow \widehat{H}$ is continuous; the kernel $H^\perp$, the inverse image of the open singleton $\{1\}$, is open. The last two statements follow from [theorems 11.2](#) and [12.2](#). $\square$

12.1 Products and sums

Proposition 12.5 (Finite products). *For LCA groups G and H , restriction to the two factors is a topological isomorphism*

$$(\widehat{G \times H}) \xrightarrow{\sim} \widehat{G} \times \widehat{H}.$$

Its inverse sends (χ, ψ) to $(g, h) \mapsto \chi(g)\psi(h)$.

Proof. Every (g, h) factors as $(g, 1)(1, h)$, which proves that the displayed maps are algebraic inverses. They are continuous by compact–open functoriality. For continuity of the inverse, if $K \subseteq G \times H$ is compact, its projections K_G, K_H are compact. Choose identity neighbourhoods $V_G, V_H \subseteq \mathbb{T}$ with $V_G V_H \subseteq V$. Then

$$W(K_G, V_G) \times W(K_H, V_H)$$

maps into $W(K, V)$. This proves continuity at the identity. $\square$

Proposition 12.6 (Discrete sums and compact products). *Let $(D_i)_{i \in I}$ be discrete abelian groups and $(K_i)_{i \in I}$ compact abelian groups. Then*

$$\widehat{\left(\bigoplus_{i \in I} D_i \right)} \simeq \prod_{i \in I} \widehat{D_i}, \quad (46)$$

$$\widehat{\left(\prod_{i \in I} K_i \right)} \simeq \bigoplus_{i \in I} \widehat{K_i}. \quad (47)$$

The first isomorphism is between compact groups, and the second between discrete groups.

Proof. A homomorphism from the algebraic direct sum is uniquely a family of homomorphisms from the factors, and continuity is automatic because the sum is discrete. Compact subsets of a discrete space are finite, so the compact–open topology is pointwise convergence on finitely many elements; this is exactly the product topology. This proves (46).

For the second formula, a finite-support family of characters defines a character of the product. Conversely, let $\chi : \prod_i K_i \rightarrow \mathbb{T}$ be continuous. The inverse image of $A_{\pi/2}$ contains a basic identity neighbourhood $\prod_i U_i$, with $U_i = K_i$ outside a finite set F . For $j \notin F$, the entire coordinate subgroup K_j maps into $A_{\pi/2}$; by [theorem 2.1](#), it maps trivially. Hence χ depends on only the coordinates in F . This proves algebraic bijectivity. Both sides are discrete: the product is compact, so its dual is discrete, and every $\widehat{K_i}$ is discrete and the direct sum is given the discrete topology. Thus the bijection is a homeomorphism. $\square$

12.2 A general Fourier inversion criterion

Theorem 12.7. *Let $f \in L^1(G)$. If $\widehat{f} \in L^1(\widehat{G})$, then*

$$f(x) = \int_{\widehat{G}} \widehat{f}(\chi) \chi(x) d\chi \quad (48)$$

for almost every x . The right side is a continuous function vanishing at infinity; hence (48) holds at every point at which f has that continuous representative. In particular, the Fourier transform is injective on $L^1(G)$.

Proof. Let $g(x) = \int \widehat{f}(\chi) \chi(x) d\chi$. Dominated convergence and the Riemann–Lebesgue argument on the dual group show $g \in C_0(G)$. With e_U from (19), Fubini and the already proved inversion formula for e_U give

$$(f * e_U)(x) = \int_{\widehat{G}} \widehat{f}(\chi) \widehat{e_U}(\chi) \chi(x) d\chi.$$

The left side converges to f in L^1 . Since $|\widehat{e_U}| \leq 1$ and $\widehat{e_U} \rightarrow 1$ pointwise, dominated convergence shows that the right side converges to g uniformly in x . On each compact $K \subseteq G$, uniform convergence implies convergence in $L^1(K)$, while global L^1 convergence also implies convergence in $L^1(K)$. Uniqueness of the local L^1 limit gives $f = g$ almost everywhere on every compact set; Radon inner regularity then gives equality almost everywhere on G . If $\widehat{f} = 0$, then $g = 0$, proving injectivity. $\square$

13 Canonical examples

In the following examples additive notation is used inside formulas, while the abstract duality functor remains multiplicative.

Example 13.1 ($\mathbb{Z}$ and the circle). The map

$$\mathbb{T} \longrightarrow \widehat{\mathbb{Z}}, \quad z \longmapsto (n \mapsto z^n)$$

is a topological isomorphism. Indeed, a homomorphism from $\mathbb{Z}$ is uniquely determined by the image of 1, and every such homomorphism is continuous because $\mathbb{Z}$ is discrete. The displayed map is continuous; it is a continuous bijection from compact $\mathbb{T}$ to the Hausdorff space $\widehat{\mathbb{Z}}$, hence a homeomorphism. Dualizing and applying [theorem 10.7](#) gives the canonical isomorphism

$$\widehat{\mathbb{T}} \simeq \mathbb{Z}, \quad n \longmapsto (z \mapsto z^n).$$

The topology on $\widehat{\mathbb{T}}$ is discrete by compactness of $\mathbb{T}$.

Example 13.2 (The real line). For $\xi \in \mathbb{R}$ put

$$\chi_\xi(x) = \exp(2\pi i \xi x).$$

Then $\xi \mapsto \chi_\xi$ is a topological isomorphism $\mathbb{R} \simeq \widehat{\mathbb{R}}$.

To prove surjectivity without invoking biduality, let $\chi : \mathbb{R} \rightarrow \mathbb{T}$ be a continuous additive character. On a sufficiently small interval $(-\delta, \delta)$, the image lies in an arc on which a continuous argument is unique. Write $\chi(x) = e^{2\pi i a(x)}$ there, with $a(0) = 0$. Whenever $x, y, x+y \in (-\delta, \delta)$, the difference $a(x+y) - a(x) - a(y)$ is an integer and varies continuously; at $(0, 0)$ it is zero, so it is zero throughout the connected small domain. Thus a is locally additive. For arbitrary x , choose n with $x/n \in (-\delta, \delta)$ and set $\tilde{a}(x) = na(x/n)$. Local additivity shows that this is independent of n and is a continuous additive lift of χ . Every continuous additive $\tilde{a} : \mathbb{R} \rightarrow \mathbb{R}$ has the form $\tilde{a}(x) = \xi x$: first for rational x , then for real x by density. Hence $\chi = \chi_\xi$.

The parameter map is continuous for the compact-open topology by uniform continuity of $(\xi, x) \mapsto e^{2\pi i \xi x}$ on compact rectangles. Conversely, uniform closeness of χ_ξ to 1 on $[-1, 1]$ first forces $|\xi| < 1/2$ (otherwise some $x \in [-1, 1]$ maps to -1), after which the principal argument at $x = 1$ controls $|\xi|$. Thus the inverse parameter map is continuous.

Example 13.3 (Euclidean groups). By [theorems 12.5](#) and [13.2](#),

$$\widehat{\mathbb{R}^n} \simeq \mathbb{R}^n, \quad \xi \longmapsto (x \mapsto e^{2\pi i \langle \xi, x \rangle}).$$

Under Lebesgue measure on both sides this is the normalization used in [\(1\)](#); it is self-dual, so no $(2\pi)^n$ factor occurs in inversion.

Example 13.4 (Finite cyclic groups). For $C_n = \mathbb{Z}/n\mathbb{Z}$ with the discrete topology,

$$C_n \xrightarrow{\sim} \widehat{C_n}, \quad k \longmapsto \left(m \mapsto \exp\left(\frac{2\pi i k m}{n}\right) \right).$$

This is an algebraic bijection between finite discrete groups and hence a topological isomorphism. Counting measure on C_n is dual to $(1/n)$ times counting measure on its dual with our inversion convention.

Example 13.5 (Finite abelian groups). The structure theorem writes a finite abelian group noncanonically as a finite product of cyclic groups. The preceding example and finite-product duality give a noncanonical isomorphism $A \simeq \widehat{\widehat{A}}$. In contrast, the evaluation isomorphism $A \simeq \widehat{\widehat{A}}$ is canonical and natural. This distinction should be preserved in a formal API: a finite group generally has no preferred isomorphism with its first dual.

Example 13.6 (Tori and lattices). For the n -torus $\mathbb{T}^n$ and the lattice $\mathbb{Z}^n$,

$$\widehat{\mathbb{T}^n} \simeq \mathbb{Z}^n, \quad \widehat{\mathbb{Z}^n} \simeq \mathbb{T}^n.$$

More generally, if $\Lambda \leq \mathbb{R}^n$ is a lattice, quotient duality gives

$$(\widehat{\mathbb{R}^n / \Lambda}) \simeq \Lambda^\perp = \{\xi \in \mathbb{R}^n : \langle \xi, \lambda \rangle \in \mathbb{Z} \text{ for every } \lambda \in \Lambda\},$$

the usual dual lattice, with the discrete topology.

Example 13.7 (Infinite sums and products). For any set I , [theorem 12.6](#) specializes to

$$\widehat{\left(\bigoplus_I \mathbb{Z}\right)} \simeq \mathbb{T}^I, \quad \widehat{(\mathbb{T}^I)} \simeq \bigoplus_I \mathbb{Z}.$$

The first source is discrete and the first target compact; the second source is compact even for uncountable I , and every continuous character depends on finitely many coordinates.

14 A Mathlib-oriented implementation plan

This section translates the proof into declarations and dependency layers. It is not a claim that the proposed declaration names already exist. The items marked “existing” were checked against Mathlib v4.32.0; all others are proposed additions.

14.1 Existing kernel

Existing declaration or module	Role
<code>PontryaginDual</code>	Definition as <code>A ->_t* Circle</code> , with induced compact–open topology.
<code>PontryaginDual.ext</code> , <code>one_apply</code>	Extensionality and evaluation simp normal forms.
<code>PontryaginDual.map</code> , <code>map_apply</code> , <code>map_comp</code>	Contravariant precomposition.
<code>PontryaginDual.mapHom</code>	Continuity of the map operation in a locally compact codomain parameter.
<code>ContinuousMonoidHom.isInducing_toContinuousMap</code>	Transfers compact–open topology results from continuous maps.
<code>ContinuousMonoidHom.instContinuousEval</code>	Joint evaluation for a locally compact domain.
<code>ContinuousMonoidHom.continuous_of_continuous_uncurry</code>	Currying joint evaluation to construct α_G .
<code>ContinuousMonoidHom.locallyCompactSpace_of_hasBasis</code>	Local compactness of the character group from a countable arc basis in $\mathbb{T}$.
<code>Circle.centeredArc</code> , <code>Circle.mem_centeredArc</code> , <code>Circle.hasBasis_centeredArc</code> , <code>div_two_pow</code>	The small-arc basis.

Existing declaration or module	Role
<code>Circle.eq_one_of_forall_pow_mem_centeredArc_pi_div_two</code>	No small subgroups for the circle.
<code>Mathlib.MeasureTheory.Measure.Haar.Basic</code>	Haar measure and invariance.
<code>Mathlib.MeasureTheory.Group.IntegralConvolution</code> and <code>Mathlib.Analysis.Convolution</code>	Convolution infrastructure.
<code>Mathlib.Analysis.Convex.KreinMilman</code>	Krein–Milman.
<code>Mathlib.Analysis.Normed.Module.WeakDual</code>	Weak dual topology and Banach–Alaoglu infrastructure.
<code>Mathlib.Topology.ContinuousMap.StoneWeierstrass</code>	Density of the Fourier algebra.
<code>Mathlib.MeasureTheory.Integral.RieszMarkovKakutani.Basic</code>	Representation of positive functionals by Radon measures.

The source already gives the dual three central instances: `T2Space`, `IsTopologicalGroup`, and `LocallyCompactSpace`. It also gives compactness for a discrete source and discreteness for a compact source. These facts should be reused, not reproved inside the biduality file.

14.2 Recommended file and declaration layers

Layer	Suggested module	Principal new declarations
L0	<code>Topology/Algebra/PontryaginDual/Evaluation</code>	<code>eval</code> , <code>eval_apply</code> , group-hom laws, continuity, naturality, triangle identity.
L1	<code>Analysis/Fourier/LCA/ApproximateIdentity</code>	Normalized C_c bumps, translation continuity in L^p , approximate identities, compact-uniform convergence of their character integrals.
L2	<code>Analysis/Fourier/LCA/Spectrum</code>	LCA convolution algebra, Riemann–Lebesgue, L^1 spectrum equivalence, transform topology equals compact–open, density in C_0 , Fourier–Stieltjes uniqueness.
L3	<code>Analysis/Fourier/LCA/PositiveType</code>	Finite-matrix positive type, integral criterion, elementary bounds, convolution-square examples.
L4	<code>Analysis/Fourier/LCA/GNS</code>	Semidefinite quotient, Hilbert completion, strongly continuous cyclic representation, extreme/irreducible criterion, abelian Schur lemma.
L5	<code>Analysis/Fourier/LCA/Bochner</code>	Weak-* compact positive ball, Krein–Milman proof, representing positive measure, uniqueness.

Layer	Suggested module	Principal new declarations
L6	Analysis/Fourier/LCA/ Inversion	$V^1(G)$, σ_f , multiplier class $\mathcal{F}$, dual Haar measure, inversion and scaling.
L7	Analysis/Fourier/LCA/ Plancherel	Dense-domain Parseval, unitary extension, L^2 product-convolution, localized Fourier transform.
L8	Topology/Algebra/ PontryaginDual/Duality	Injectivity, neighbourhood-basis recovery, closed image, surjectivity, <code>ContinuousMulequiv</code> biduality.
L9	Topology/Algebra/ PontryaginDual/Exact	Annihilators, quotient dual, double annihilator, character extension, strict exactness.
L10	Topology/Algebra/ PontryaginDual/Examples	$\mathbb{Z}$, $\mathbb{T}$, $\mathbb{R}$, finite cyclic groups, finite products, discrete sums/compact products.

This layering keeps the existing lightweight definition file from importing measure theory and functional analysis. In particular, the definition and functorial map remain usable without importing the proof of biduality.

14.3 Evaluation map skeleton

The first implementation milestone is independent of the analytic proof. A typical constructor skeleton is:

```
namespace PontryaginDual

noncomputable def eval (G : Type*)
  [CommGroup G] [TopologicalSpace G]
  [IsTopologicalGroup G] [LocallyCompactSpace G] :
  G →_* PontryaginDual (PontryaginDual G) where
toFun x :=
  { toFun := fun chi => chi x
    map_one' := by simp
    map_mul' := by intro chi psi; rfl
    continuous_toFun := continuous_eval_const x }
map_one' := by ext chi; exact chi.map_one
map_mul' x y := by ext chi; exact chi.map_mul x y
continuous_toFun := by
  apply ContinuousMonoidHom.continuous_of_continuous_uncurry
  -- discharge with joint evaluation, after swapping factors
  exact ...

@[simp] theorem eval_apply (x : G) (chi : PontryaginDual G) :
  eval G x chi = chi x := rfl

end PontryaginDual
```

The actual names of the evaluation lemmas should be resolved with `#check` in the target branch. It is preferable to prove continuity through the existing `ContinuousEval` instance rather than unfold the compact-open topology.

14.4 Suggested theorem inventory

The following inventory is fine-grained enough for autoformalization. Each line should be a declaration with only earlier lines as substantive dependencies.

- P1.** Normalized C_c bump subordinate to an identity neighbourhood.
- P2.** L^p translation isometry and continuity.
- P3.** Approximate identity convergence; uniform character convergence on compact sets.
- P4.** L^1 convolution norm, associativity, involution, and convolution square of positive type.
- P5.** Fourier norm bound, continuity, translation/involution/convolution identities.
- P6.** Riemann–Lebesgue vanishing at infinity.
- P7.** Every nonzero multiplicative functional on the unitization of L^1 is bounded.
- P8.** Spectrum classification $\Delta(L^1(G)) \simeq \widehat{G}$.
- P9.** Joint evaluation for the transform topology and equality with compact–open.
- P10.** Stone–Weierstrass density of $\{\widehat{f}\}$ in $C_0(\widehat{G})$.
- P11.** Fourier–Stieltjes uniqueness in the asymmetric form of [theorem 5.4](#).
- P12.** Equivalence of finite-matrix and integral positive type.
- P13.** GNS construction and continuous cyclic coefficient representative.
- P14.** Extreme coefficient if and only if irreducible GNS representation.
- P15.** Abelian Schur lemma and classification of extreme points.
- P16.** Weak-* compactness of $P(G)$.
- P17.** Bochner existence and uniqueness.
- P18.** Symmetric measure identity [\(18\)](#).
- P19.** Positive approximate net e_U and recovery formula [\(22\)](#).
- P20.** Multiplier class closure, translation, and inclusion of $C_c(\widehat{G})$.
- P21.** Construction, nontriviality, and invariance of dual Haar measure.
- P22.** Inversion on V^1 and uniqueness of dual normalization.
- P23.** Dense Parseval isometry and density of its range.
- P24.** Plancherel and localized-transform lemma.
- P25.** Evaluation injective.
- P26.** Polar sets recover the original neighbourhood filter.

P27. Evaluation image closed.

P28. Evaluation image surjective.

P29. Assemble `ContinuousMuleEquiv`; prove naturality simp lemmas.

P30. Quotient duality, double annihilator, character extension, and exactness.

14.5 Representation choices and known pressure points

Multiplicative versus additive APIs. The public theorem should first be proved for the multiplicative `PontryaginDual`. Fourier integration uses the coercion `Circle -> Complex`. Only after the multiplicative development is stable should additive wrappers be generated or proved. Premature `to_additive` use is likely to create brittle names around convolution and character evaluation.

Positive type. Use the finite-matrix predicate as the public definition: it is algebraic, does not mention Haar measure, and applies to arbitrary topological groups. Add the integral criterion as a theorem under LCA and Haar hypotheses. For matrices indexed by `Fin n`, the expression can be written as a quadratic form; this makes the 2×2 bound and closure under limits reusable.

Complex measures. Positive measures can use `Measure`; linear combinations in $V(G)$ require complex vector measures (the `MeasureTheory.VectorMeasure` hierarchy) or an explicit four-positive-measure representation. The latter is initially easier for Bochner but makes uniqueness and multiplier identities verbose. A thin local structure containing a complex vector measure together with finite-variation evidence is likely the cleanest interface.

Weak-* topology. State positivity on L^∞ classes by the family of inequalities $\operatorname{re}\langle \varphi, k_f \rangle \geq 0$ and $\operatorname{im}\langle \varphi, k_f \rangle = 0$. These are manifestly weak-* closed. Choose continuous representatives only through the GNS theorem. Do not put `Continuous` inside the weak-* closed set.

No σ -compactness shortcut. Haar measure on an arbitrary LCA group need not be σ -finite. Any use of $(L^1)^* = L^\infty$, extraction of almost-everywhere convergent subsequences, or sequential compactness must be checked for hidden countability assumptions. The proof above uses nets, local regularity, and compact testing. If the available L^1 duality theorem is too restrictive, the spectrum proof can use the translation-ratio argument directly and avoid representing a functional by an L^∞ function.

A.e. representatives. Convolution of two L^2 functions and inverse transforms have canonical continuous representatives. Give these representatives named definitions and simp lemmas; do not silently rewrite an L^p class as a pointwise function. Every upgrade from almost everywhere to everywhere should cite continuity and full support explicitly.

Quotients. The compact-lifting lemma used in [theorem 11.2](#) deserves its own general topological-group declaration: a compact subset of G/H is contained in the image of a compact subset of G . It avoids any local-section or second-countability assumption.

Noncomputability and normalization. Haar measures, Riesz representatives, and the inverse of the evaluation bijection are noncomputable. Keep the topological homomorphism `eval` computable in its formula, but mark the assembled equivalence and analytic constructions noncomputable. Store the dual Haar measure together with its inversion property so that rescaling lemmas can use uniqueness rather than unfold choices.

14.6 No-circularity audit

Result	Why it does not presuppose Pontryagin duality
Local compactness of $\widehat{G}$	Arzelà–Ascoli plus the no-small-subgroups property of $\mathbb{T}$.
L^1 spectrum	Translation in the convolution algebra; no assertion that characters of $\widehat{G}$ are evaluations.
Fourier–Stieltjes uniqueness	Stone–Weierstrass density of transforms of $L^1(G)$; tests only evaluation functions.
Bochner	Krein–Milman and the classification of extreme positive coefficients; uses Fourier–Stieltjes uniqueness only.
Dual Haar measure and inversion	Bochner, the multiplier class, Riesz–Markov, and Haar uniqueness.
Injectivity of α_G	The already established inversion formula and a compactly supported convolution square.
Closed image	Topological embedding and the dense-locally-compact-subspace lemma.
Surjectivity of α_G	Plancherel supplies a transform supported off the closed image; asymmetric Fourier–Stieltjes uniqueness forces it to vanish.
Character extension	Proved only after biduality, by dualizing the induced quotient map; it is not an input to the analytic proof.

This audit should be mirrored in the Lean import graph. In particular, `Bochner.lean`, `Inversion.lean`, and `Plancherel.lean` must not import `PontryaginDual/Duality.lean`.

15 Dependency summary

The logical spine of the proof is:

1. compact–open characters form an LCA group and joint evaluation is continuous;
2. the L^1 spectrum is $\widehat{G}$, so Fourier transforms are dense in $C_0(\widehat{G})$ and finite measures are determined by evaluation characters;
3. GNS plus abelian irreducibility identifies extreme normalized positive-type functions with characters;
4. Banach–Alaoglu, Krein–Milman, and measure uniqueness prove Bochner’s theorem;
5. the multiplier construction produces the uniquely normalized Haar measure on $\widehat{G}$ and Fourier inversion on convolution squares;
6. inversion separates points and recovers the original topology from compact polars;

7. Plancherel produces localized Fourier transforms;
8. a localized transform outside the closed evaluation image contradicts measure uniqueness, proving surjectivity;
9. biduality then gives annihilator duality, character extension, exactness, and the categorical anti-equivalence.

Every arrow in this chain points forward. The proof therefore supplies an autoformalization order as well as a mathematical proof.

A Notation-to-Lean dictionary

Mathematics	Lean target	Comment
$\mathbb{T}$	<code>Circle</code>	Subtype of $\mathbb{C}$ of norm one; multiplicative.
$\widehat{G}$	<code>PontryaginDualG</code>	Definitional equality with <code>G ->_t* Circle</code> .
$\widehat{f}$	<code>PontryaginDual.mapf</code>	Precomposition; reverses arrows.
α_G	<code>proposed PontryaginDual.evalG</code>	Continuous homomorphism into the double dual.
$\alpha_G : G \simeq \widehat{\widehat{G}}$	<code>proposed PontryaginDual.toDoubleDualG</code>	A <code>ContinuousMulEquiv</code> ; name subject to library convention.
$W(K, U)$	<code>compact-open MapsTo set</code>	Use <code>ContinuousMap.isOpen_setOf_mapsTo</code> .
Haar measure	<code>MeasureTheory.Measure.IsHaarMeasure</code> / library Haar choice	Keep the initial measure explicit during Fourier normalization.
$f * g$	group convolution	Be explicit about the selected continuous representative.
f^*	involution $x \mapsto \overline{f(x^{-1})}$	Abelian groups are unimodular.
finite complex measure	complex <code>VectorMeasure</code> of finite variation	Positive special case is an ordinary <code>Measure</code> .
$S^\perp$	<code>proposed PontryaginDual.annihilatorS</code>	Prefer a closed subgroup when S is a subgroup.
G/H	quotient group type	Requires H closed for Hausdorff/LCA instances.

B Foundational lemmas worth packaging separately

The proof repeatedly uses the following generic statements. If any is absent in the required generality, it should be added outside the Pontryagin namespace.

- (A1) A dense locally compact subspace of a Hausdorff space is open ([theorem 10.4](#)).
- (A2) A locally compact subgroup of a Hausdorff topological group is closed. This follows by applying (A1) to its closure.

- (A3) Every compact subset of a quotient topological group G/H is contained in the image of a compact subset of G , when G is locally compact.
- (A4) A bounded continuous homomorphism $G \rightarrow \mathbb{C}^\times$ has image in $\mathbb{T}$.
- (A5) A positive functional on $C_c(X)$ is bounded on functions with support in a fixed compact set.
- (A6) If two continuous functions on a full-support measure space agree almost everywhere, they agree everywhere.
- (A7) If a continuous density u satisfies $u\mu \geq 0$ and μ has full support, then $u \geq 0$ pointwise.
- (A8) The convolution of two L^2 functions has a canonical continuous bounded representative, with the L^2 norm product bound.
- (A9) Uniform convergence on compact sets of characters, combined with an integrable tail estimate, gives continuity of the Fourier transform.
- (A10) A positive semidefinite sesquilinear form obeys Cauchy–Schwarz and descends to an inner product on the quotient by its nullspace.

C Bibliographic notes

The analytic organization follows the classical Haar–Fourier–positive-type route. The proof has been rewritten here with conventions and dependency boundaries chosen for **Mathlib**. Useful references for comparison are:

References

- [1] E. Hewitt and K. A. Ross, *Abstract Harmonic Analysis, Vol. I*, Springer, second edition, 1979.
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